What Schools Tell Families: Website Signals in U.S. K–12 School Markets*

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Abstract

The U.S. K–12 market enrolls 50 million students across 130,000 schools with no required disclosure of price or program. We measure what 81,109 public, private, and charter schools tell families on their websites, using a fully replicable layer of keyword, structural, and transparency features extracted from 1.84 million deduplicated pages. Three findings. First, every market-facing content dimension responds positively to local same-sector competitor density except religious identity, which is flat across 1–30 km on the private side ($\hat{\beta} = -0.005$, *n.s.* at 10 km) and consistent with affiliation-anchored signaling. Second, on the private side a website mention of financial aid is associated with 1.77 percentage points faster annual enrollment growth ($\hat{\beta} = +0.018$, $p < 0.001$) and 41% lower school-closure odds; academic-rigor and tuition-transparency content carry the same direction. Third, the academic-rigor coefficient on growth *flips sign* in the public sector ($\hat{\beta} = -0.0013$, $p < 0.001$), ruling out generic website-quality effects in favour of sector-specific sorting on observable content.

Work in Progress, May 2026. This draft is built on the national live-website crawl (~940,000 unique pages across ~81,000 K–12 schools, both public and private) and uses keyword and structural measures as the primary measurement layer. The condensed two-stage LLM ontology pipeline is in Appendix B as a validation exercise on a 2,958-school subsample.

1 Introduction

Only 39 percent of U.S. private K–12 schools post a tuition amount anywhere on their website. Among public schools the share is 3 percent; among charter schools it is 4 percent. The U.S. K–12 market enrolls more than 50 million students across roughly 130,000 schools, but no regulator requires the most basic price information to be disclosed. State report cards cover public schools unevenly and exclude private schools. What a family can learn about a school before enrolling is what the school chooses to put on its website.

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Schools make those choices deliberately. A Catholic parish school in Louisiana leads with scripture and parish identity. A charter school in Phoenix leads with robotics and a dual-language program. A district school in suburban New Jersey leads with sports schedules and bus routes. The choices map to the families each school depends on, and the stakes are real: private-school closures have run at roughly 3 percent per year over the past decade, and public-school enrollment redistribution across districts and charter alternatives has accelerated.

This paper asks what U.S. K–12 schools say on their websites, why they say it, and what those choices predict for enrollment and survival. We crawl 81,109 school websites — 66,725 public schools (including 4,769 charter schools) and 14,384 private schools — and recover 1.84 million deduplicated pages. We extract a deterministic measurement layer: structural features, page-type presence, transparency signals, and prevalence-weighted scores on 22 keyword topic families. The headline finding has three parts. First, every market-facing content dimension except religious identity rises with local same-sector competitor density; religious identity is flat. Second, the strongest private-side predictor of enrollment growth is whether a school's website mentions financial aid: schools that do grow 1.8 percentage points faster per year than schools that do not, controlling for state, affiliation, locale, and lagged enrollment. Third, the same financial-aid mention and academic-rigor keyword density together predict a roughly 40 percent reduction in closure odds in a six-state private-school panel.

The descriptive cross-section is sharply sector-specific. Private-school websites dominate every market-facing transparency signal: 39 percent list a tuition amount (public: 3 percent; charter: 4 percent), 61 percent mention financial aid (public: 30 percent), and 47 percent mention accreditation (public: 15 percent). Religious identity keyword density appears on 70 percent of private-school websites, 24 percent of traditional public, and 21 percent of charter. Charter schools sit between private and traditional public on most market-facing dimensions and lead both on market-urgency language ("apply now," "limited seats": charter 65 percent vs. private 41 percent and traditional public 29 percent) and on individualized-learning vocabulary (charter 36 percent, traditional public 15 percent). The cross-sector contrast establishes that website content carries sector-specific information that researcher-assigned categories alone do not.

Local market structure shapes content choices in predictable ways with one clean exception. Religious identity is flat across competitor density on the private side ($\hat{\beta} = -0.005$, $n.s.$ at 10 km), consistent with identity-anchored demand that does not respond to block-level alternatives. Every other focal topic — family partnership, tuition transparency, market urgency, individualized learning, academic rigor — has a positive coefficient on log same-sector competitor count, with magnitudes of 0.02–0.04 standard deviations per log-unit of competitors. The public side shows positive gradients of similar sign but 3×–4× larger magnitudes. The exception — religious identity — isolates the single dimension that markets do not treat as a competitive content margin.

Enrollment growth and school closure respond to website content. In the private panel of 38,692 school-years across 12,437 schools, tuition transparency keyword density and academic-rigor keyword density predict faster annual growth ($\hat{\beta} = +0.006$ and $+0.005$, both $p < 0.001$), and the financial-aid mention flag adds 1.8 percentage points. On the public side the same academic-rigor measure flips sign ($\hat{\beta} = -0.0013$, $p < 0.001$ in a 251,792 school-year panel): public schools that foreground college-prep language tend to be schools facing enrollment pressure from charter and magnet alternatives. The cross-sector sign reversal rules out a generic "websites-that-look-better-grow-faster" story and points to sector-specific sorting on the same observable content dimension. In a six-state closure panel of 1,553 private schools and 70 events, the academic-rigor and financial-aid predictors return as protective signals (odds ratios 0.66 and 0.59 respectively).

The paper makes three contributions. First, the measurement: a national, cross-sector, fully replicable website-content measurement on 81,109 K–12 schools, built from deterministic structural and keyword features that any researcher can reproduce from a fixed dictionary. We supplement this with a condensed two-stage LLM ontology measurement on a 2,958-school subsample (Appendix B); the two measurement layers agree on the broad patterns but diverge on specific findings in ways we report transparently. Second, the cross-sector design. Prior work studies either the public sector (where test-score and report-card disclosure exists) or specific private segments (charter, Catholic). We measure public, private, and charter schools on the same content primitives and document where their signaling logic converges and where it diverges. Third, the substantive results connect website content — a school's own self-presentation — to enrollment growth and school survival, the two market outcomes that matter most for school operators and families.

We extend a recent literature on school-choice signals, information frictions, and market structure that has studied test-score disclosure (**?**), charter and voucher competition (**??**), and school-website marketing in specific contexts (**???**). Our contribution to that literature is the cross-sector measurement and the direct linkage to enrollment growth and closure on the same unit of observation. Section 2 develops the connection.

Section 3 describes the corpus and measurement layer. Section 4 reports the descriptive cross-section. Section 5 reports the competition and community-context gradients. Section 6 reports the enrollment-growth and closure results. Section 7 discusses implications and the two extensions in progress: a within-school Internet Archive panel that fixes the cleanest identification gap, and a national closure-panel expansion to all 50 states. Appendix B reports the condensed LLM-ontology validation including the unsupervised topic-model check requested by coauthor review.

2 Related Literature

This paper sits at the intersection of three literatures: school choice and information frictions, identity and organizational-niche theories of school markets, and text-as-data measurement of firm self-presentation. Each has a piece of what we want; none has measured the school-controlled signal layer at national, cross-sector scale.

The school-choice literature has established that families respond to information about schools when that information is made legible. **?** show that simplified test-score disclosure shifts families toward higher-scoring schools in Charlotte; **?**, **?**, **?**, and **?** document parallel responses in other settings. More recent work studies charter and voucher competition: **?** and **?** measure incumbent-school responses to new competitive entry; **?** trace demand reallocations under choice expansion; **?** link school-level features to exit in a developing-country market. A common limitation across this work is that the information environment families face is defined from the demand side — test scores, peer composition, commute times — and the school-controlled signal layer that families actually encounter first enters only as researcher-assigned categories (Catholic, charter, test-score tier). We measure that signal layer directly.

Our competition results have a theoretical home in two adjacent traditions. **?**'s identity economics predicts that religious-school families derive direct utility from a school that matches their identity, so religious signaling should be institutionally anchored in affiliation rather than tactically responsive to local competition. **?**'s organizational-niche framework predicts that operationally costly specialist signals — high-touch family-community claims, individualized pedagogy — are credible in thin markets where small scale can deliver them, and fade in denser markets where school size and standardization rise. The flat religious-identity competition gradient we document on the private side is direct evidence for the identity-economics prediction. The keyword-density measurement does not cleanly reproduce the LLM-prominence finding of declining family-partnership signal with competitor density; we report both layers and discuss the divergence in Section 5 and Appendix B.7. The literature on the economics of religion — **??**, more recently — argues that identity-bounded organizations command more inelastic demand and higher durability, which provides the underlying mechanism for our closure result that mission-forward identity content predicts lower closure odds.

The papers closest to our object of study come from outside mainstream economics and focus directly on school websites and mission statements. **?** show that charter-school websites function as sorting devices by signaling fit to specific families. **?** uses a national charter-school website corpus to study how projected school identities track demographic stratification. **?** recover school purposes from mission statements; **?** studies how market-facing schools produce consumer-oriented self-information. These papers establish that school self-presentation is substantive and measurable. We differ in three ways. We apply the logic to the full cross-sector universe rather than to charter schools alone or to mission statements alone. We measure on common content primitives that are fully replicable from a fixed keyword dictionary, so the same measurement can be applied to historical Internet Archive snapshots without rerunning expensive inference. And we link the content layer directly to enrollment growth and closure on the same unit of observation, so the measurement is anchored to market outcomes that matter.

The substantive education-economics literature on private and religious schooling provides the relevant baseline for our private-side results. **?**, **?**, and **?** document the Catholic-school effect on educational attainment and its survival under selection-on-unobservables corrections; **?** supply a theoretical framework for private–public competition. **?** documents measurable outcome differences for Montessori pedagogy, which connects to our finding that individualized-learning vocabulary tracks alternative-school affiliation more than demand-side appeal. **?** provides the industrial-organization framework for firm survival under costly differentiation that our closure result implicitly applies.

The methodology connects to the text-as-data tradition in economics. **?** survey the methods; **?** demonstrate the approach by measuring product differentiation from SEC filings; **?** extend the toolkit to firm-level text features. The central methodological lesson of that literature is that large-scale text becomes credible economic data when the coding scheme is disciplined, validated externally, and connected to outcomes through transparent designs. School websites fit naturally: they are public, school-controlled, directed at the audience that drives demand, and produced specifically to shape family beliefs about the school. That intentionality distinguishes them from many text corpora used in economics, where the author's purpose is not to influence the outcome being measured. A school website is, in the language of signaling theory, a costly signal the school chooses to send. Its content is an economic decision.

We make one further methodological choice that bears explicit discussion. The main text uses deterministic keyword and structural features rather than LLM-extracted ontology scores. The choice trades off finer-grained prominence measurement against replicability and scale. Keyword features are reproducible from a fixed dictionary, costless at national scale, extensible to historical snapshots without re-running inference, and verifiable by external researchers. LLM-based ontology measures recover more nuanced prominence distinctions but depend on model versions that are not stable across vendors, and applying them to the full ∼940,000-page national corpus at sustained inference cost is not feasible in the current iteration. We retain the LLM ontology exercise on a 2,958-school subsample as a validation asset (Appendix B). Where the two measurement layers converge — religious-identity competition flatness, transparency and quality signals predictive of growth and survival — the findings are robust to measurement choice. Where they diverge, we report both and flag the substantive interpretation issue (Appendix B.7).

The remaining gap, which this paper does not close, is causal identification. The cross-sectional level of website content is a function of school identity, history, and market position. The associations we report are predictive rather than causal. A within-school panel built from Internet Archive snapshots of the same schools' websites in 2018 and 2026 provides the most credible path to identification (Section 7); that panel is data work in progress.

3 Data and Measurement

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing. The recently-collected tuition$ continuous extraction, tract-level ACS demographics, and EDFacts achievement controls are wired into the appendix robustness (Sec. D.2); they will be folded into main-text regressions in the next iteration.]

This section describes the school universe, the national website corpus, the deterministic measurement layer built on top of it, and the outcome panels used in Sections 5 and 6. Every variable in the main-text analysis is computed from page metadata, deterministic page-type rules, or keyword dictionaries. The condensed LLM ontology validation on a 2,958-school subsample appears in Appendix B.

3.1 School universe and URL coverage

The starting universe is the union of two federal registries: the NCES Common Core of Data (CCD) for public schools and the NCES Private School Survey (PSS) for private schools, with the latest available waves (CCD 2022–23; PSS 2021–22). The combined universe contains 145,570 schools across all 50 states, the District of Columbia, and U.S. territories: 100,418 public schools (including 8,288 charter schools), 22,345 private schools, and 22,807 closed or transitional schools that we exclude from the active frame.

We recover a verified seed URL for each active school through an eight-stage pipeline: school-submitted URL fields in PSS and CCD; NCES CCD directory `WEBSITE` records; OpenStreetMap, Wikidata, and Wikipedia structured data; IRS Form 990 electronic filings; public-school and private-school aggregator profiles; within-district sibling inference for public schools sharing a Local Education Agency; stratified web-search recovery; and spot-check validation. Coverage as of May 2026, restricted to schools currently classified as Open or New, reaches **93.3 percent** overall (114,531 of 122,763 active schools): 97.1 percent of active public schools (97,479 of 100,418) and 76.3 percent of active private schools (17,052 of 22,345). Table 1 reports the breakdown.

Table 1: National URL Coverage by Sector (Active Schools)

Segment	Total active	Seeded URL	% seeded
Public schools	100,418	97,479	97.1%
Private schools	22,345	17,052	76.3%
Total active	122,763	114,531	93.3%
Incl. closed/inactive	145,569	122,415	84.1%

Notes: Active means current PSS or CCD status of Open or New. "Seeded URL" means a candidate URL passed both an HTTP liveness check and a name-identity match against the school's official record. Eight-stage discovery pipeline described in Appendix A.1. Source: NCES PSS 2021–22 and CCD 2022–23 merged with school-website registry (May 2026).

The unseeded tail is unwebbed rather than missed. Three pieces of evidence support this. First, every active public school without a seed URL also has no URL in NCES's own CCD directory `WEBSITE` field. Second, unseeded active private schools enroll a median of 39 students against an active-private median of 82, with 56 percent enrolling 50 students or fewer — school sizes typical of home-based or family-run programs without a public web presence. Third, a stratified spot-check on 609 unseeded schools found 94 percent of large unseeded public schools (500+ enrollment) via independent web search (pipeline misses, now being recovered) but essentially zero among small private schools (under 100 enrollment) and zero in West Virginia (where district pages substitute for individual school sites). Appendix A.1 reports the detail.

3.2 National live-website corpus

We crawled the seeded universe in April–May 2026 using ten parallel virtual machines. The crawler fetches the homepage plus interior pages prioritized by semantic proximity to mission, about, academics, and admissions content. The first full pass capped recovery at 20 pages per school (homepage plus 19 ranked internal links). Schools that hit this initial cap entered a deeper re-crawl wave with a 50-page ceiling, on the rationale that any school exhausting the 20-page budget likely has substantive content worth retrieving.

The post-deduplication corpus contains **1,835,152 unique pages across 81,109 schools**: 66,725 public schools (61,956 traditional + 4,769 charter) and 14,384 private schools. Mean unique pages per crawled school is 22.6 (private 23.0; public 22.5); only 2.8 percent of crawled schools reach the 50-page ceiling. Mean total deduplicated word count per school is approximately 33,000. The raw page-fetch table contains 1,918,157 entries; the gap to the 1,835,152 unique count is within-school duplication (about 4.3 percent), typical of websites that expose the same content under multiple URLs.

Attrition from active-seeded to crawled. Of the 114,624 active-seeded schools, 81,109 yielded substantive crawl content, leaving a gap of 33,515 schools. These are sites whose homepage returned an HTTP response sufficient for activity classification but for which the deeper crawl produced no usable page text. The dominant failure modes, ordered by prevalence in the crawl error log, are: (i) Cloudflare and anti-bot 403 walls that pass the initial homepage ping but block subsequent requests; (ii) JavaScript-only sites whose content does not materialise in either static fetch or Playwright headless rendering; and (iii) sites that return only soft-404 stubs, generic landing pages, or login walls beyond the homepage. Active-but-uncrawled schools skew toward small private schools on hosted-platform domains and toward charter networks behind Cloudflare; section 3.7 reports the public/private split and the state distribution. We treat the 81,109 crawled schools as the analysis universe throughout, with explicit checks for selection on observables in Appendix A.1.

Shared domains. A large share of crawled schools live on websites that host multiple sibling schools: 18.5 percent of private schools (diocesan platforms, multi-campus academies) and 66.3 percent of public schools (district websites). The relevant question — raised in pre-submission review by Ben Arold — is whether content fetched from a shared domain actually represents the individual school or only the parent organisation. We address this in two ways. First, in a stratified sample of 6,000 schools (1,500 per sector $\times$ shared/solo cell) we verify that each school's own name appears in its retrieved text. The distinctive non-stop token from the school name matches in 95 percent of solo-domain schools (both sectors), 91 percent of shared-domain private schools, and 84 percent of shared-domain public schools (Table 2). The shared-vs-solo gap of $\approx$11 percentage points on the public side is the main residual concern; it implies that for roughly one in six shared-domain public schools, the crawler did not surface text that mentions the school by name and may be measuring district-level rather than school-specific content. Second, every main-text private-side regression is re-estimated with an `on_shared_domain` dummy as a sensitivity check; Table 30 in Appendix D.1 shows that all four headline topic coefficients are unchanged to four decimal places and the dummy itself is indistinguishable from zero.

Table 2: School-name validation by sector and domain status

Sector	Domain	N (sample)	Full-name match (%)	Distinctive-token match (%)
Private	Solo	1,500	68.6	95.1
Private	Shared	1,500	55.0	91.0
Public	Solo	1,500	65.9	95.3
Public	Shared	1,500	43.3	84.1

Notes: For each crawled school we concatenate page text (title, headings, body, normalised) and check whether (a) the full normalised school name and (b) the longest non-stop distinctive token appear as a substring. "Solo" = the school is the only one on its primary domain; "Shared" = at least one sibling school shares the domain. The distinctive-token rate is the relevant lower bound on whether the retrieved text references the school by name.

Table 3: Headline Corpus Counts

Quantity	Value	Definition
Seeded schools	122,415	Schools in the national frame (NCES public + PSS private) for which we attempted URL discovery.
Active seeded	114,624	Seeded schools with a live, reachable homepage at crawl time (others were closed, redirected to a district page, or unreachable).
Crawled schools	81,109	Active-seeded schools from which we successfully retrieved at least one page of substantive content.
Unique pages retrieved	1,835,152	Total unique pages across all crawled schools, after within-school deduplication.
Raw pages crawled	1,918,157	Total pages fetched before dedup; the gap to unique is duplicated boilerplate (about 4.3%).
Mean unique pages/school	22.6	After-dedup; per-school cap was 20 on the initial pass plus a deeper re-crawl wave for cap-hitting sites (effective ceiling ≈50).
Coverage vs. seeded	66.3%	Crawled schools as a share of the full seeded frame (81,109 / 122,415).
Coverage vs. active seeded	70.8%	Crawled schools as a share of schools with a live homepage — the rate conditional on reachability.
Hit per-school cap	2,309 (2.8%)	Crawled schools whose unique-page count reached 50 (the effective ceiling after the re-crawl wave).
Active but uncrawled	33,515	Active-seeded schools with no successful page retrieval. The dominant failure modes are Cloudflare / anti-bot 403 walls, JavaScript-only sites that resisted both static and headless Playwright fetch, and pages returning only soft-404 stubs or login walls.
Uncrawled seeded (total)	41,306	Sum of inactive seeds (homepage unreachable) and active-but-uncrawled seeds.

Notes: Seeded schools are schools in the national website registry passed to the live crawl. Crawled schools have at least one recovered usable page in the deduplicated corpus. Per-school cap: 20 pages on the initial crawl plus a deeper re-crawl wave (50-page ceiling) for schools that exhausted the initial budget; 2.8 percent of crawled schools reach the 50-page ceiling. Source: `features_national_20260517_v3.csv`.

3.3 Measurement layer

For every crawled school we compute four families of features and write them to a single school-level feature file (`features_national_20260517_v3.csv`, 81,109 rows). The construction script (`scripts/84ca_keyword_structural_extractor.py`) is fully replicable from the deduplicated page text plus a fixed keyword dictionary (Appendix A.3).

Tier 1: structural features. Pages crawled, unique pages, maximum crawl depth, total deduplicated word count, mean words per page, soft-404 flags, low-content flags, homepage indicator. These capture website investment and basic scale.

Page-type presence. Binary flags for admissions, tuition, academics, mission/about, athletics, arts, staff, news/blog, calendar, virtual-tour, library, and policy/handbook pages. Detection rules combine URL-path keywords, title text, and heading text.

Transparency and digital-infrastructure signals. `tuition_amount_present` (regex match on dollar values within keyword windows), `financial_aid_present`, `accreditation_mentioned`, `test_scores_listed`, social-media platform count (Facebook, Instagram, YouTube, X, TikTok, LinkedIn), multilingual content, embedded video, online application portal, parent portal, PDF-link presence.

Tier 2: keyword-topic scores. Twenty-two topic-family dictionaries applied to normalized page text. Each topic produces three measures: a binary presence flag (any school page with ≥ 1 keyword hit), a raw count of hits, and a prominence-weighted score that weights homepage hits $\times 3$, heading hits $\times 2$, body hits $\times 1$, and deeper-page hits $\times 0.5$. Families include religious identity, character and civic formation, family partnership, individualized and project-based learning, academic rigor and college prep, market urgency, tuition transparency, STEM, arts and athletics, DEI, sustainability, SEL and wellness, and others listed in Appendix A.3.

Table 4: Measurement Menu: Tier-1 Structural and Tier-2 Keyword Families

Measure family	Examples	Status	Comment
Website investment	Pages, depth, words, cap-hit status	Ready now	Current crawl; lower bounds for capped schools
Page inventory	Admissions, tuition, academics, policy, parent portal	Ready now	Improves after deeper crawl
Digital infrastructure	Social links, PDF use, video, multilingual signals	Ready now	Current crawl
Transparency	Tuition amounts, financial aid, accreditation, test scores	Ready now	Best refreshed after fuller extraction
Navigation access	Admissions, tuition, contact, parent, and academic links	Ready now	Uses navigation labels, headings, and page titles
Keyword families	Academic, identity, market, family, community topics	Ready now	Current crawl; prominence improves after deeper crawl
Historical change	Added or removed language over time	Needs Wayback text	Not yet stable
Topic models / embeddings	Bottom-up clustering and distinctiveness	Needs extra modeling	Separate build step

Notes: Tier 1 measures are deterministic structural features; Tier 2 are deterministic keyword-topic prevalence scores. No LLM inference enters any main-text variable. Dictionary listed in Appendix A.3.

3.4 Why deterministic measurement

The deterministic layer offers three advantages for cross-sector measurement at national scale. First, replicability: any researcher with the keyword dictionary and the extraction script can reproduce every feature on any corpus. Second, scale: the full national pass costs effectively nothing in inference spend, so the same measurement applies to a historical Internet Archive panel without rerunning expensive LLM inference. Third, transparency: families, school operators, and regulators can inspect the dictionary and verify which words drive which scores.

The cost is granularity. Keyword counts treat "God" and "faith" as equivalent contributions to religious-identity prominence. They cannot distinguish "we welcome students of all faiths" from "we are a faith-based community." Prominence in the homepage-vs-deep-page sense is captured only through the weighting scheme, not through page-level pragmatic context. To check that the deterministic layer recovers the substantive structure a model-based layer would, we ran a two-stage LLM ontology pipeline on a 2,958-school subsample. Appendix B reports the validation: keyword and ontology measures correlate at $r \approx +0.4$ to $+0.7$ on their natural-pair topics; the dominant family identified by each measure agrees on roughly half of schools; and the broad descriptive and outcome patterns survive the substitution. Where the two layers disagree on specific findings — family-partnership competition gradient, growth-side individualized-learning vs. academic-rigor, within-Nonsectarian closure protection — we report both and discuss the divergence transparently (Appendix B.7).

3.5 Outcome panels

Two outcome panels link the measurement layer to administrative records.

Enrollment growth. For private schools we use PSS biennial enrollment from waves 2013–14, 2015–16, 2017–18, 2019–20, and 2021–22, constructing annualized log growth (half of biennial log growth) at each transition. This produces 38,692 school-year observations across 12,437 private schools. For public schools we use the NCES Common Core of Data Membership file for six waves (2016–17, 2017–18, 2018–19, 2020–21, 2021–22, 2022–23; the 2019–20 file is currently unreadable in our environment and is omitted). The public panel contains 251,792 school-year observations across 65,735 public schools.

School closure. We assemble a state Department of Education closure panel from official enrollment rosters in six states: California, New York, Pennsylvania, Illinois, Ohio, and Wisconsin. CA, NY, PA, and IL extend to a 2024–25 endpoint; OH and WI to 2023–24. Closure is defined as permanent disappearance from the state DOE roster in year $t + 1$, persisting in subsequent years. The current panel contains 1,553 matched private schools and 70 observed closure events. PA, IL, and OH 2024–25 files are next on the data agenda and add roughly 38 events; an all-state extension is the more substantial data project flagged in Section 7.

3.6 Summary statistics

Table 5 reports summary statistics for the main estimation samples by segment (private, public traditional, public charter). The descriptive cross-sectional patterns are striking enough to merit a full section; we defer the discussion to Section 4.

Table 5: Headline Sample Characteristics by Segment

	Private	Public traditional	Public charter
N schools	14,384	61,956	4,769
Website investment (per-school cap = 50 unique pages)			
Raw pages retrieved (mean)	23.8	23.7	22.1
Unique pages after within-school dedup (mean)	23.0	22.7	21.1
Total word count, deduplicated (mean)	33,140.1	34,450.3	29,711.9
Keyword topic prevalence (% with any hit)			
Religious identity	70.0%	23.8%	21.3%
Family partnership	64.0%	64.5%	67.5%
Individualized learning	36.2%	14.8%	35.8%
Academic rigor / college prep	78.1%	76.8%	81.3%
Tuition transparency	82.8%	49.8%	68.4%
Market urgency / admissions	40.7%	29.4%	65.4%
STEM	73.1%	67.1%	72.5%
Arts / athletics	78.6%	76.6%	77.1%
DEI / inclusion	49.2%	43.3%	54.5%
Sustainability	56.8%	50.3%	55.2%
SEL / wellness	58.3%	63.0%	68.7%
Character / civic	91.8%	85.0%	88.3%
Structural/transparency flags (% present)			
Tuition amount (regex $)	38.8%	3.4%	4.1%
Financial aid mentioned	60.6%	30.4%	34.8%
Accreditation mentioned	47.2%	14.7%	24.8%
Test scores listed	4.4%	3.0%	3.9%
Page-type presence (% with ≥ 1 page)			
Mission / about page	79.1%	58.8%	68.6%
Admissions page	60.0%	32.8%	51.3%
Tuition page	35.8%	2.5%	3.0%
Athletics page	21.3%	16.2%	12.8%
Arts page	16.1%	8.5%	9.0%
News / blog	32.3%	24.2%	30.6%
Policy / handbook	21.0%	28.1%	26.7%

Notes: Unit of observation: school. Three segments shown: private (PSS, n = 14,384), public traditional (CCD non-charter, n = 61,956), public charter (CCD charter, n = 4,769). Website-investment rows: mean values. All other rows: percentage of crawled schools in segment with the listed signal present. Source: `features_national_20260517_v3.csv` merged with the national school master. Charter classification from the CCD `is_charter` field.

3.7 Coverage limits and the historical extension

Two coverage limits bear explicit mention. First, the corpus reflects the 50-page effective ceiling after the deeper re-crawl wave; only 2.8 percent of crawled schools reach this ceiling, so the upper-tail truncation is mild. The larger concern is the 33,515 active-seeded schools that returned no crawlable content (Section 3.2). These schools are disproportionately small private schools on hosted platforms (Wix, Squarespace, Weebly) that rely on client-side rendering and Cloudflare bot protection. We report the public/private split in Appendix A.1 and run selection-on-observables checks; population-frame topic-prevalence numbers in the descriptive section should be read as lower bounds on the share of schools that produce any matching content.

Second, the Wayback historical panel that supports the cleanest identification strategy — within-school change in messaging between 2018 and 2026 — currently covers 13,718 schools with at least one Internet Archive snapshot, of which 7,930 have a snapshot in calendar years 2017 or 2018, the intended pre-treatment window. The current iteration of this paper uses this panel for a pilot within-school change regression in Appendix C; the headline cross-sectional regressions remain population-based, and the within-school panel is the identification cross-check.

Common Crawl as a future extension. A separate historical layer based on Common Crawl is technically feasible but yields essentially no usable URLs for our school-domain set. We piloted a distributed Common Crawl CDX sweep on a 5-VM fleet in May 2026 and recovered 46,233 query checkpoints covering 17,784 schools across six CC-MAIN crawl epochs. Zero of the 17,784 schools had any CC CDX hit: 99.8 percent of queries returned an explicit error from `index.commoncrawl.org`, and the remaining 0.2 percent returned empty result sets. The proximate cause was the well-known unreliability of the public CC CDX HTTP endpoint (a single AWS EC2 instance with no SLA); the deeper cause is that Common Crawl's seed list prioritises large-domain coverage and rarely visits small institutional websites. A proper implementation using AWS Athena over Common Crawl's columnar S3 index would bypass the HTTP endpoint at a one-off cost of $10–$50 in scan fees and would resolve the reliability problem, but on present evidence it would not change the empirical conclusion: Common Crawl does not have the school-website coverage our panel needs. The current paper therefore relies on the Internet Archive Wayback panel (Appendix A.6) for any historical-coverage statements, and we record the CC null result here for completeness. See `notes/cc_panel_pending_20260520.md` for the full handoff.

4 Descriptive Patterns in School Self-Presentation

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing. The recently-collected tuition$ continuous extraction, tract-level ACS demographics, and EDFacts achievement controls are wired into the appendix robustness (Sec. D.2); they will be folded into main-text regressions in the next iteration.]

This section reports what U.S. K–12 schools put on their websites, organized in three steps: the cross-sector contrast (where private, traditional public, and charter schools diverge), the geographic pattern (where each topic concentrates), and the community-context correlates of content emphasis. All numbers in this section come from `features_national_20260517_v3.csv` unless noted otherwise.

4.1 Private, traditional public, and charter schools occupy different content layers

The three segments diverge sharply on transparency signals, mildly on academic-quality signals, and not at all on operational topics. Three patterns organize the cross-sector contrast and Table 5 shows them at full breakdown.

Private-school websites dominate every market-facing transparency signal. Roughly 39 percent of private schools list a tuition amount on their website against 3.4 percent of traditional public and 4.1 percent of charter schools. Financial-aid mentions follow the same pattern (61 vs. 30 vs. 35 percent); accreditation mentions follow it more sharply (47 vs. 15 vs. 25 percent). Table 6 reports the headline pooled-public numbers; the full charter breakdown is in Table 21 in Appendix A.2. The private sector's transparency lead is not driven by website scale: mean total word count per school is roughly equal across segments (Table 7). Private schools allocate their text to disclosing the information families need to choose them.

Table 6: Transparency Signals by Sector

Signal	Private	Public
Tuition amount listed (regex match)	38.8%	3.4%
Financial aid mentioned	60.6%	30.7%
Accreditation mentioned	47.2%	15.4%
Test scores listed	4.4%	3.1%

Notes: Percentage of crawled schools with the named deterministic transparency signal detected on at least one recovered page. Sample: 14,384 private schools and 66,725 public schools (pooled traditional + charter). The charter-vs.-traditional split is in Table 21. Source: `features_national_20260517_v3.csv`.

Table 7: Website Investment by Sector

Statistic	Private	Public
Raw pages retrieved (mean)	23.8	23.6
Unique pages after dedup (mean)	23.0	22.5
Word count, deduplicated (mean)	33,140	34,112
Mean max crawl depth	1.7	1.6
Schools (N)	14,384	66,725

Notes: Means across crawled schools by sector. Page-recovery counts are exposed to the first-pass six-page cap and should be read as lower bounds on true site scale. Source: `features_national_20260517_v3.csv`.

Charter schools sit between private and traditional public on most market-facing dimensions, and on two they exceed private. Charter schools post admissions pages at 51 percent (private 60 percent, traditional public 33 percent), but they lead both other segments on market-urgency keyword density — "apply now," "limited seats," "deadline" — with 65 percent prevalence against 41 percent private and 29 percent traditional public. They also nearly match private schools on individualized-learning vocabulary (36 percent charter vs. 36 percent private; only 15 percent in traditional public). Charters are behaving as alternative-pedagogy public schools with an active enrollment funnel.

Religious-identity content remains a private-sector phenomenon. Religious keywords appear on 70 percent of private-school websites against 24 percent of traditional public and 21 percent of charter schools. The charter share matches the traditional-public share, indicating that the small charter share of religious-language schools is plausibly the same religiously-affiliated public academies in dense urban markets that show up in traditional-public data. Character and civic vocabulary, by contrast, is near-universal across all three segments (88–92 percent), which is the measurement issue we flagged for the closure regression in Section 6.4: broad character vocabulary lacks the within-group variation needed for closure identification.

Table 8 reports the top keyword topics by sector, ranked by the union of each sector's top ten. The page-type inventory in Table 9 confirms the cross-sector contrast at the structural level: private schools are roughly twice as likely as public schools to have a mission, admissions, or tuition page reachable from the homepage.

Table 8: Top Keyword Topic Prevalence by Sector

Topic	Private	Public
Character / values	91.8%	85.2%
Tuition transparency	82.8%	51.2%
Arts	78.6%	76.6%
College prep / rigor	78.1%	77.1%
STEM	73.1%	67.5%
Denominational	71.0%	16.1%
Religious identity	70.0%	23.6%
Family partnership	64.0%	64.8%
SEL / wellness	58.3%	63.4%
Alumni	58.2%	45.3%
Athletics	47.7%	63.0%
Learning support	40.6%	62.8%
Technology	46.4%	50.9%

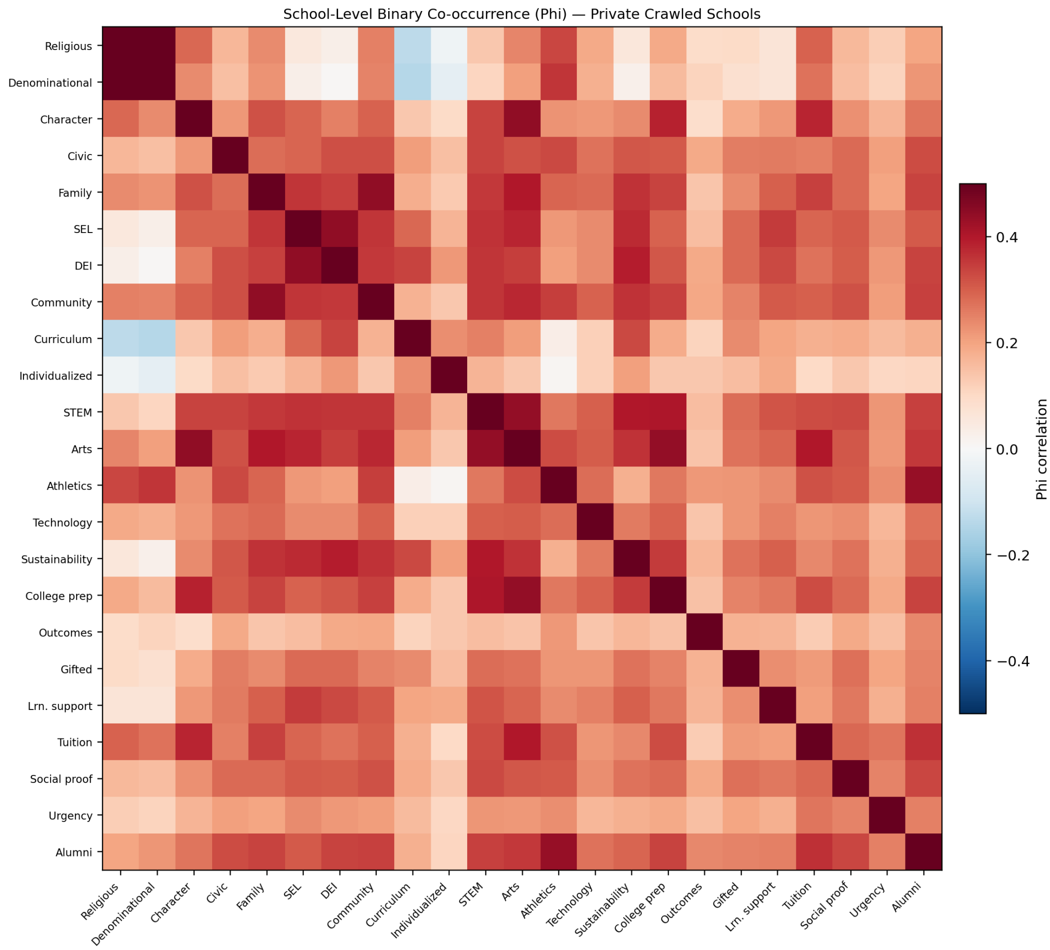
Notes: Percentage of crawled schools in each sector with at least one page hitting the named keyword family. Topics are the union of each sector's top-ten by prevalence. Source: `features_national_20260517_v3.csv`.

Table 9: Page-Type Inventory by Sector

Page type	Private	Public
Mission / about	79.1%	59.5%
Admissions	60.0%	34.1%
Tuition	35.8%	2.5%
Academics	54.7%	43.0%
Athletics	21.3%	15.9%
Arts	16.1%	8.5%
Staff / faculty	52.7%	42.6%
News / blog	32.3%	24.6%
Policy / handbook	21.0%	28.0%
Calendar	53.2%	43.5%

Notes: Percentage of crawled schools in each sector with at least one recovered page classified into the listed page type. Page-type classification uses deterministic URL-path, title, and heading rules described in Appendix A.4. Lower bounds under the first-pass six-page crawl cap.

Figure 1: Topic Co-Occurrence Across School Websites

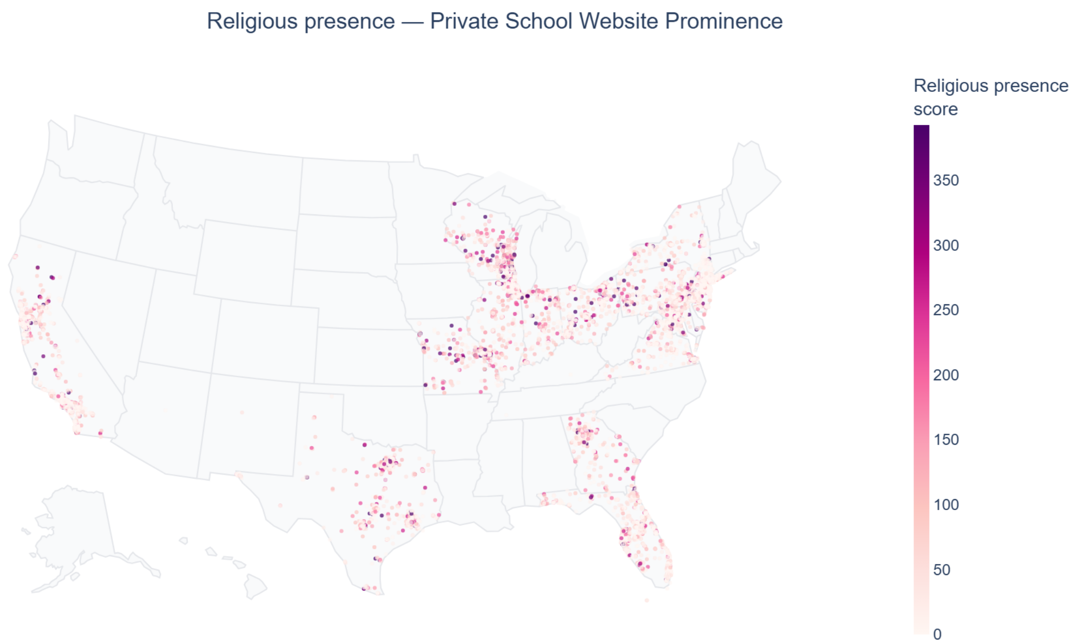


Notes: Pearson phi correlations among binary topic-presence indicators across n = 14,384 crawled private schools. Color scale runs from dark blue (negative co-occurrence) to dark red (positive co-occurrence). The clearest off-diagonal pattern is the negative correlation between religious identity and individualized-learning topics — schools emphasizing religious identity tend to de-emphasize individualized pedagogy, and vice versa. Most other off-diagonal cells are near zero.

4.2 Geography: where each topic concentrates

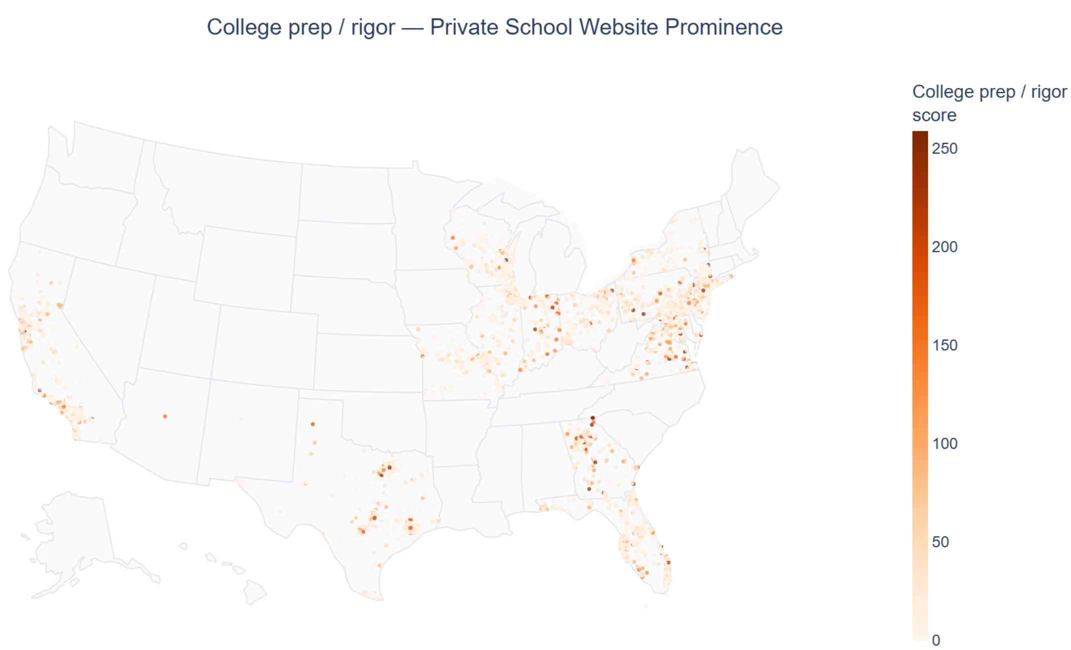
The cross-sector contrast in Section 4.1 runs alongside sharp geographic concentration of specific content. Figure 2 maps religious-identity keyword density at the school level; the Southeast and Appalachian states show the highest concentrations, the Pacific Northwest and New England the lowest. Figure 3 maps academic rigor / college-prep keyword density; this content concentrates in metropolitan areas across the Northeast and West Coast and on a set of identifiable preparatory-school clusters in the Mid-Atlantic and inland California. Figure 4 maps tuition-transparency keyword density and shows the inverse pattern of religious-identity density: transparency concentrates where religious affiliation does not. Three additional maps appear in Appendix A.8: STEM, character/civic, and parent-involvement density at the school level.

Figure 2: Religious-Identity Keyword Density, U.S. Private K–12 Schools



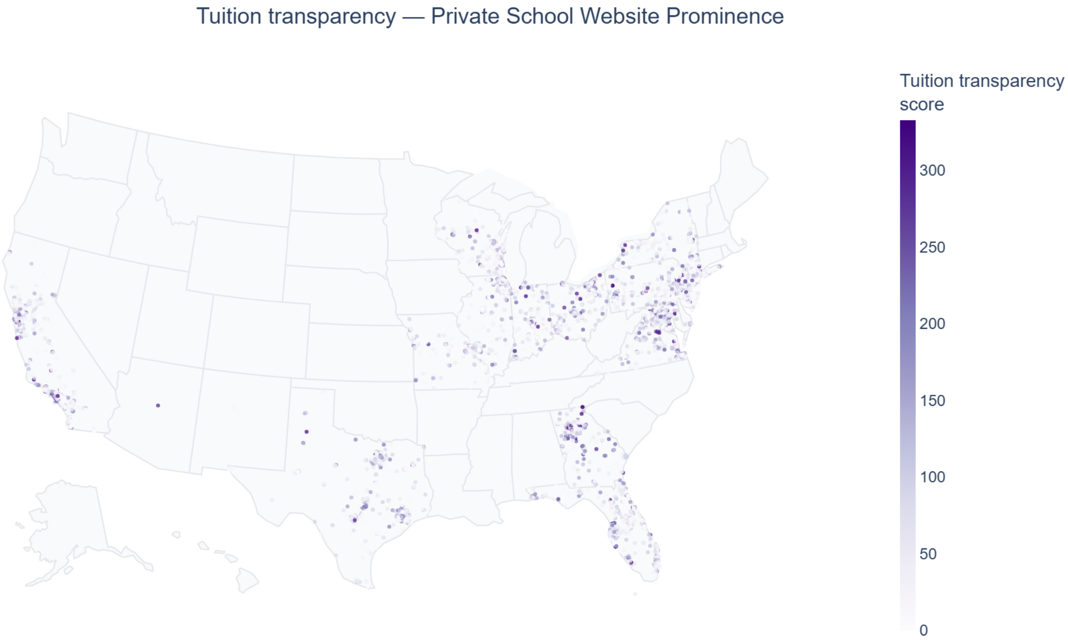
Notes: Sample: n = 14,384 private K–12 schools with crawled websites. Each dot is one school; color intensity is the prominence-weighted religious-identity keyword score. The highest densities concentrate in the Southeast and Appalachian states; the lowest run along the Pacific Northwest and New England.

Figure 3: Academic Rigor / College-Prep Keyword Density, U.S. Private K–12 Schools



Notes: Sample: n = 14,384 private K–12 schools. Each dot is one school; color intensity is the prominence-weighted college-prep keyword score. Concentration in major Northeast and West Coast metropolitan areas reflects the urban-private prep cluster.

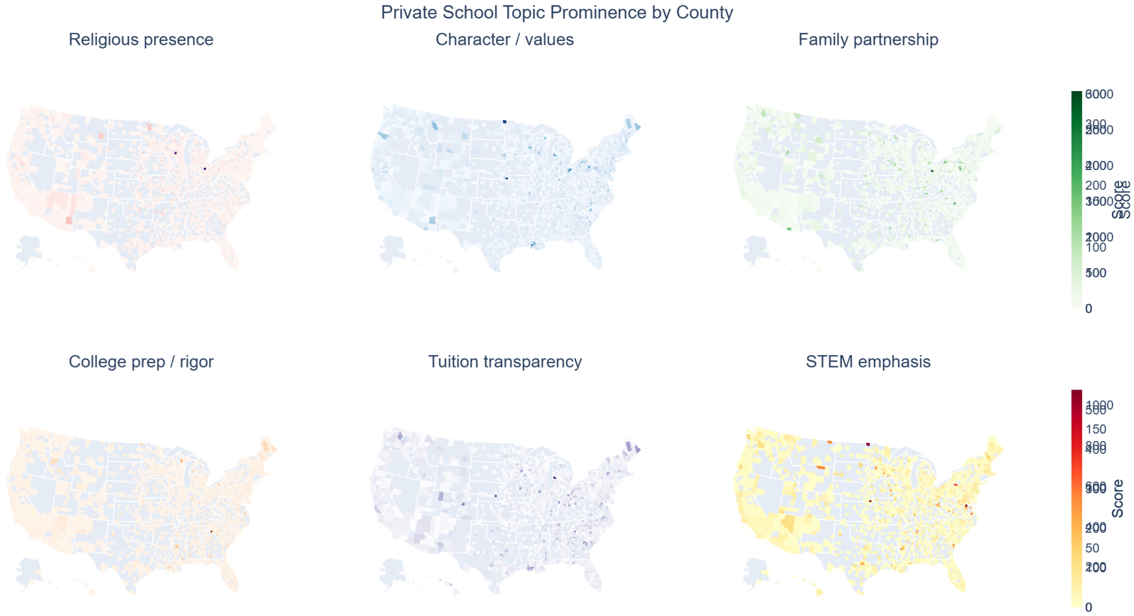
Figure 4: Tuition-Transparency Keyword Density, U.S. Private K–12 Schools



Notes: Sample: $n = 14{,}384$ private K–12 schools. Each dot is one school; color intensity is the prominence-weighted tuition-transparency keyword score. The high-density regions are roughly the inverse of the religious-identity map: tuition disclosure concentrates where religious affiliation is sparse.

A county-level choropleth at Figure 5 aggregates these dot-level patterns into county shares and reveals two substantive regularities at the county level. Counties with high religious-identity shares cluster along a corridor from the Texas Panhandle through Kentucky and the Appalachian South. Counties with high college-prep shares cluster in the New York–Boston, San Francisco–San Jose, and Washington–Northern Virginia corridors.

Figure 5: County-Level Topic Prominence: Six-Panel Choropleth



Notes: Six-panel U.S. county choropleth of mean private-school keyword-topic prominence, one panel per focal topic (top row: religious presence, character/values, family partnership; bottom row: college prep / rigor, tuition transparency, STEM emphasis). Each panel uses its own color scale; scores are not comparable across topics. Counties with no crawled private school are blank. Religious presence concentrates in the South-Central Bible Belt and Appalachian South; college-prep concentrates in Northeast Corridor metros and inland California.

4.3 State-level structure

State-level aggregation supports an interpretive check: which states are where on each focal dimension. Figure 6 reports the state-mean prevalence of three focal topics — religious identity, academic rigor, and tuition transparency — and orders states from highest to lowest religious-identity score. The vertical ranking is intuitive: at the top of the religious-identity list sit Mississippi, Alabama, Louisiana, Arkansas, and Tennessee. At the bottom sit Oregon, Washington, Massachusetts, and Vermont. The academic-rigor ordering is roughly orthogonal to the religious-identity ordering, consistent with the negative co-occurrence in Figure 1.

Figure 6: Top-20 States by Topic Prominence, Six Focal Topics

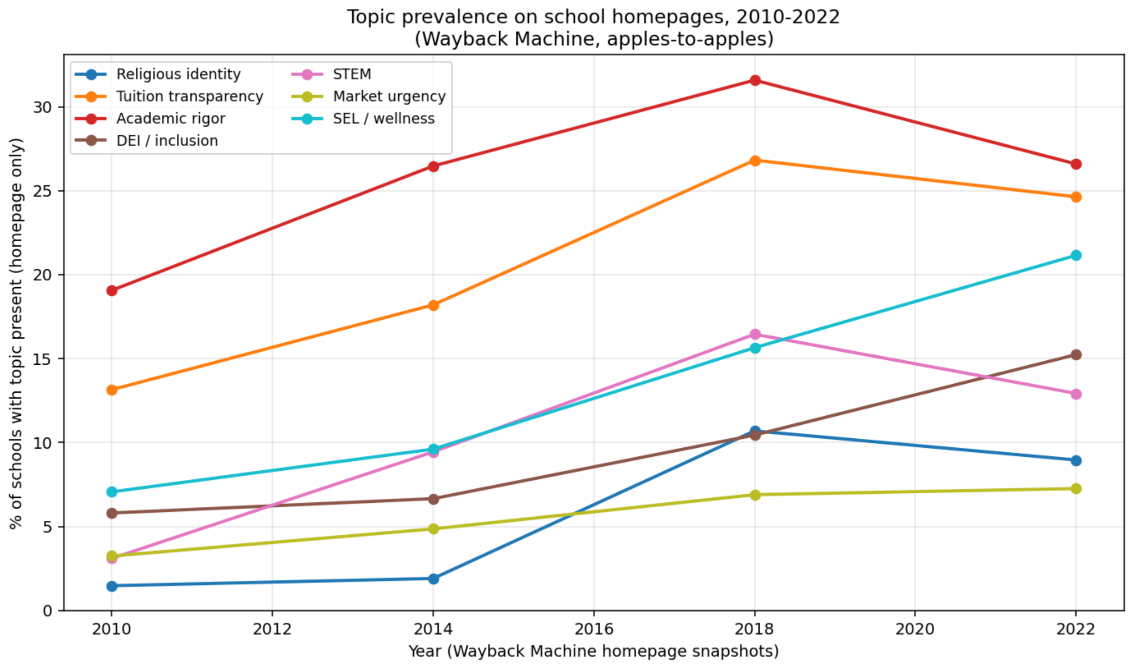
Notes: Six-panel horizontal bar chart. Each panel shows the top 20 states by mean private-school keyword-topic prominence, one panel per focal topic (top row: religious presence, character/values, family partnership; bottom row: college prep / rigor, tuition transparency, STEM emphasis). State two-letter codes on the y-axis; mean score on the x-axis. Panels use independent x-axis scales; rankings are not directly comparable across topics.

4.4 How school self-presentation has changed over time

[editorial: Numbers in this subsection refresh as the multi-year Wayback sweep (currently ≈28% complete; 5 Hetzner VMs running) and Common Crawl quarterly sweep (currently 48 of 89 epochs complete) finish.]

The 2026 cross-section is one snapshot of a moving target. Pairing the Wayback Machine homepage-only panel across 2010, 2014, 2018, and 2022 with the 2026 deep crawl reveals which topics have entered, exited, or held steady on school websites over a decade and a half. To keep the comparison apples-to-apples we restrict the 2010–2022 series to Wayback homepage-only features, which is what Wayback consistently captures; the 2026 deep crawl is separated visually because its higher per-school page count mechanically inflates topic counts.

Figure 7: Topic prevalence on school homepages, 2010–2022 (Wayback Machine)

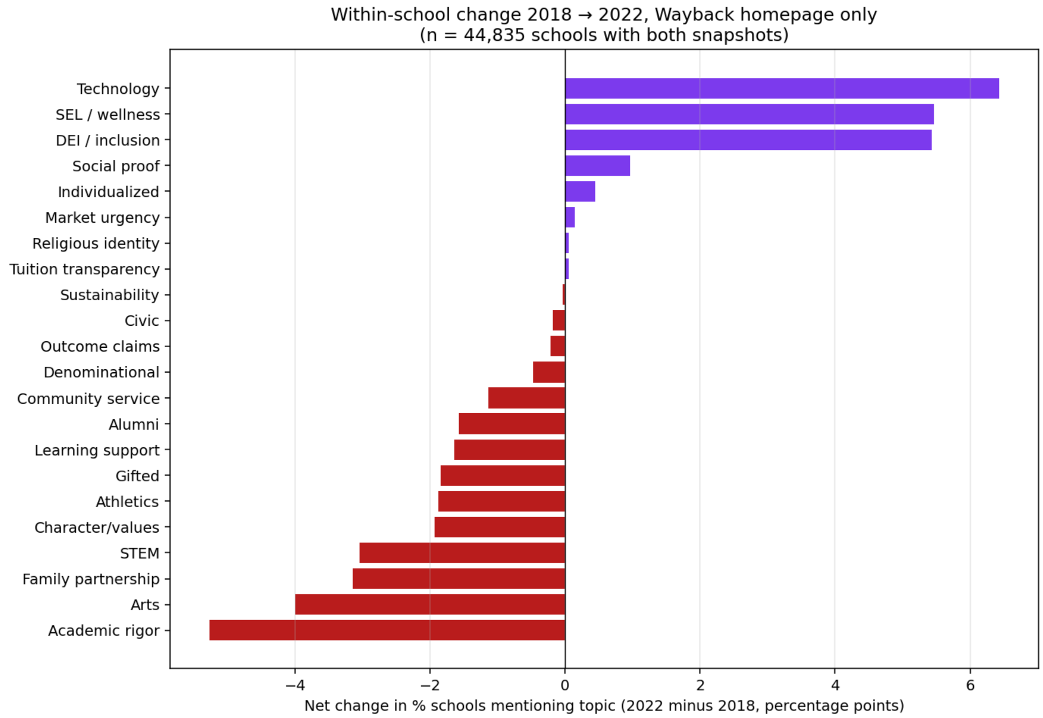


Notes: Cross-sectional means by year: the share of crawled schools whose homepage contains any keyword for each topic family. Wayback Machine homepage snapshots only, to make years comparable. Sample size varies by year as Wayback coverage of small institutional websites is sparser in earlier years (2010: 3,765 schools; 2014: 5,398; 2018: 30,566; 2022: 10,017 so far with the multi-year sweep still running). The 2026 deep-crawl panel reaches substantially higher topic-coverage shares because it indexes 20+ pages per school rather than just the homepage and is therefore omitted from this chart; within-school 2018→2022 changes (Figure 22) use homepage-only on both ends and are interpretable.

Three patterns are visible. First, religious-identity keyword density on homepages rose from 1.5% (2010) to 10.7% (2018), then held essentially constant through 2022 (9.0%) — consistent with the cross-sectional finding that religious self-presentation is anchored to school affiliation rather than to time-varying market conditions. Second, tuition transparency and academic-rigor language followed similar inverted-U paths: rapid rise through 2018, then a slight pull-back. Third, the post-pandemic period shows clearly distinct trajectories for SEL/wellness (7% → 21%) and DEI/inclusion (5.8% → 15.2%): these two grew substantially even within the same set of schools.

Within-school 2018 → 2022. Restricting to the 44,835 schools we observe in both Wayback 2018 and Wayback 2022 (homepage only), Figure 22 shows that the time pattern is not driven by composition: within the same school panel, the same topics moved. **Technology (+6.4 pp), SEL/wellness (+5.5 pp), and DEI/inclusion (+5.4 pp) are the topics most often *added* by schools between 2018 and 2022.** Academic rigor (−5.3 pp), arts (−4.0), family partnership (−3.1), STEM (−3.1), athletics (−1.9), and character/values (−1.9) were *dropped* more often than they were added. **Religious identity stayed essentially fixed within school (**+0.1 **pp net, 2.1% added, 2.0% dropped)** — one of only two topics with near-zero churn, again consistent with affiliation-anchored signaling. Tuition transparency was similarly stable within school (+0.0 pp net).

Figure 8: Within-school change in topic prevalence, 2018 → 2022 (Wayback homepage)



Notes: Net change in topic-presence rate, 2022 minus 2018, for the 44,835 schools with both a 2018 and a 2022 Wayback homepage snapshot. Both years use homepage-only features so the comparison is apples-to-apples. Topics in red were dropped more often than they were added; topics in purple were added more often. The full add/drop breakdown is in Table 29.

Table 10: Within-school topic change 2018 → 2022 (Wayback homepage)

Topic	2018 (%)	2022 (%)	Net (pp)	Added (pp)	Dropped (pp)
Academic rigor	32.000000	26.700000	-5.300000	11.200000	16.400000
Arts	34.500000	30.500000	-4.000000	12.900000	16.900000
Family partnership	33.000000	29.800000	-3.100000	11.500000	14.700000
STEM	15.900000	12.900000	-3.100000	7.200000	10.200000
Character/values	46.100000	44.200000	-1.900000	15.900000	17.800000
Athletics	45.300000	43.400000	-1.900000	10.600000	12.500000
Gifted	17.300000	15.500000	-1.800000	7.300000	9.100000
Learning support	27.900000	26.200000	-1.600000	10.000000	11.700000
Alumni	27.900000	26.300000	-1.600000	9.700000	11.300000
Community service	7.500000	6.400000	-1.100000	3.300000	4.400000
Denominational	11.200000	10.800000	-0.500000	2.100000	2.600000
Outcome claims	1.100000	0.900000	-0.200000	0.600000	0.800000
Civic	7.900000	7.700000	-0.200000	4.300000	4.500000
Sustainability	12.200000	12.200000	-0.000000	7.100000	7.100000
Tuition transparency	25.700000	25.800000	0.000000	10.700000	10.700000
Religious identity	9.300000	9.300000	0.100000	2.100000	2.000000
Market urgency	7.000000	7.100000	0.100000	5.100000	5.000000
Individualized	5.000000	5.400000	0.400000	1.900000	1.500000
Social proof	8.600000	9.600000	1.000000	6.100000	5.200000
DEI / inclusion	10.200000	15.600000	5.400000	10.300000	4.800000
SEL / wellness	15.800000	21.200000	5.500000	13.200000	7.700000
Technology	10.800000	17.200000	6.400000	11.800000	5.300000

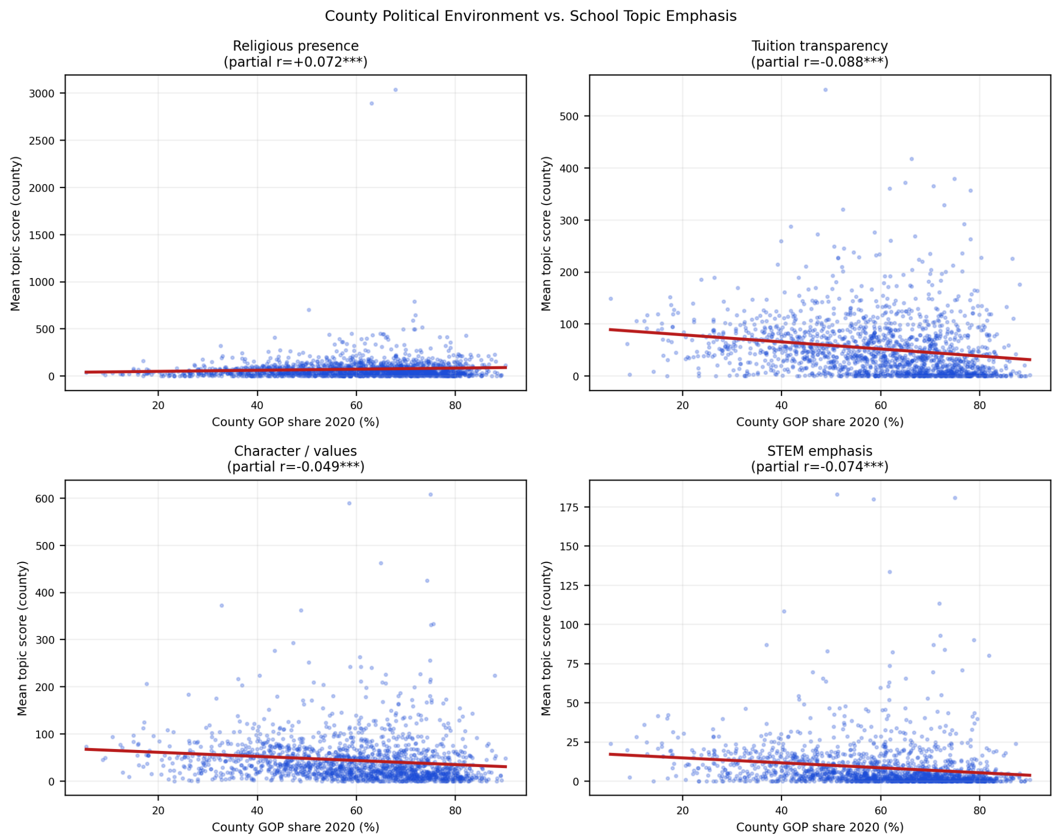
Notes: Per-topic decomposition. "Added (pp)" is the share of schools where the topic was absent in 2018 but present in 2022; "Dropped (pp)" is the opposite. "Net (pp)" is the difference. Sample: 44,835 schools with Wayback homepage features in both years.

The interpretation is straightforward: schools update their websites with the topics that the current educational discourse foregrounds (post-2018 attention to mental health, equity, and technology) while letting older emphases recede on the homepage. The exception is religious identity, which is fixed at the school-affiliation level rather than the content-marketing level.

4.5 Community context

The descriptive section closes with the community-context patterns that motivate the determinants analysis in Section 5. Religious-identity keyword density correlates strongly with the county Republican vote share ($r = +0.28$, $p < 0.001$) and negatively with the share of county adults holding a bachelor's degree ($r = -0.27$, $p < 0.001$). Both correlations survive within-state demeaning (partial $r = +0.25$ and -0.25 respectively), ruling out the possibility that the patterns reflect cross-state composition alone. Individualized-learning keyword density follows the opposite pattern: positive with educational attainment, negative with Republican vote share. Market-interface signals — admissions-page presence, tuition transparency — show no significant correlation with any community-context variable. Schools diverge sharply on identity and pedagogy along community lines but converge on how explicitly they present practical enrollment mechanics.

Figure 9: County-Level Voting Correlates of Topic Prominence



Notes: Scatter plots of county-mean topic prominence against county-level 2020 Republican two-party vote share, for religious identity, individualized learning, and tuition transparency. Religious identity correlates strongly positively; individualized learning correlates negatively; tuition transparency is roughly flat.

These descriptive patterns set up the two empirical questions of the rest of the paper. Section 5 asks whether the variation we observe responds to local market structure as well as to community context. Section 6 asks whether the same content predicts the two market outcomes that matter most for schools: enrollment growth and school closure.

5 Determinants of School-Website Content

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing. The recently-collected tuition$ continuous extraction, tract-level ACS demographics, and EDFacts achievement controls are wired into the appendix robustness (Sec. D.2); they will be folded into main-text regressions in the next iteration.]

The descriptive cross-section in Section 4 shows that schools vary systematically in what they put on their websites. This section asks what drives that variation. We focus on two candidate drivers — local same-sector competitor density and county-level community context — and hold the consequence-side question for Section 6.

5.1 Local market structure

Schools surrounded by more same-sector competitors face a stronger incentive to differentiate. The theoretical prediction splits the focal topics into two groups. Religious identity should be unresponsive to local competition: it is anchored in affiliation and community structure, and the identity-economics demand it serves is largely inelastic to block-level alternatives (**??**). Every other focal topic is candidate-differentiation content and should respond to competitor density, with the direction depending on whether the signal functions as a specialist claim (negative response: harder to deliver at scale) or as a content-investment margin (positive response: more pages, more content, on every dimension).

Specification

For each topic family k we estimate

$$\text{TopicScore}_{ik} = \alpha + \beta_k \log(\text{Competitors}_i^r + 1) + \mathbf{X}_i'\gamma + \varepsilon_i,$$

separately for private and public schools, and separately at each integer radius r from 1 to 30 km. The competitor count is drawn from the full sector universe (22,345 PSS private schools and 102,157 EDGE-geocoded public schools), not from the crawled subsample, so a school's competitive environment reflects real local density. Controls include state fixed effects, affiliation or school-type fixed effects, and log enrollment; standard errors are HC1-robust. We report selected radii in tables and the full radius curves in figures.

Private-sector results

Religious-identity keyword density is flat across every radius from 1 to 30 km ($\hat{\beta} = -0.005$, SE $= 0.007$ at 10 km; the radius curve in Figure 10 sits on the zero reference line). The flatness replicates the LLM-ontology finding from the archived draft and provides the cleanest available evidence for the identity-economics prediction: a school's religious self-presentation responds to affiliation, not to block-level competition.

Every other focal topic has a positive private-side competition gradient. At 10 km, family partnership carries $\hat{\beta} = +0.036$ (SE $= 0.007$, $p < 0.001$), tuition transparency $+0.038$ (SE $= 0.007$), market urgency $+0.024$ (SE $= 0.006$), individualized learning $+0.019$ (SE $= 0.006$), and academic rigor $+0.019$ (SE $= 0.006$). Magnitudes are economically modest but precisely estimated: a one-log-unit increase in nearby competitor count (roughly tripling competitors) shifts each topic score by 0.02–0.04 standard deviations. The radius curves (Figure 10) show that the gradients are roughly flat across radii rather than concentrated at short distances; private schools in denser markets produce more content on every market-facing dimension, and the effect is felt to roughly 20 km out.

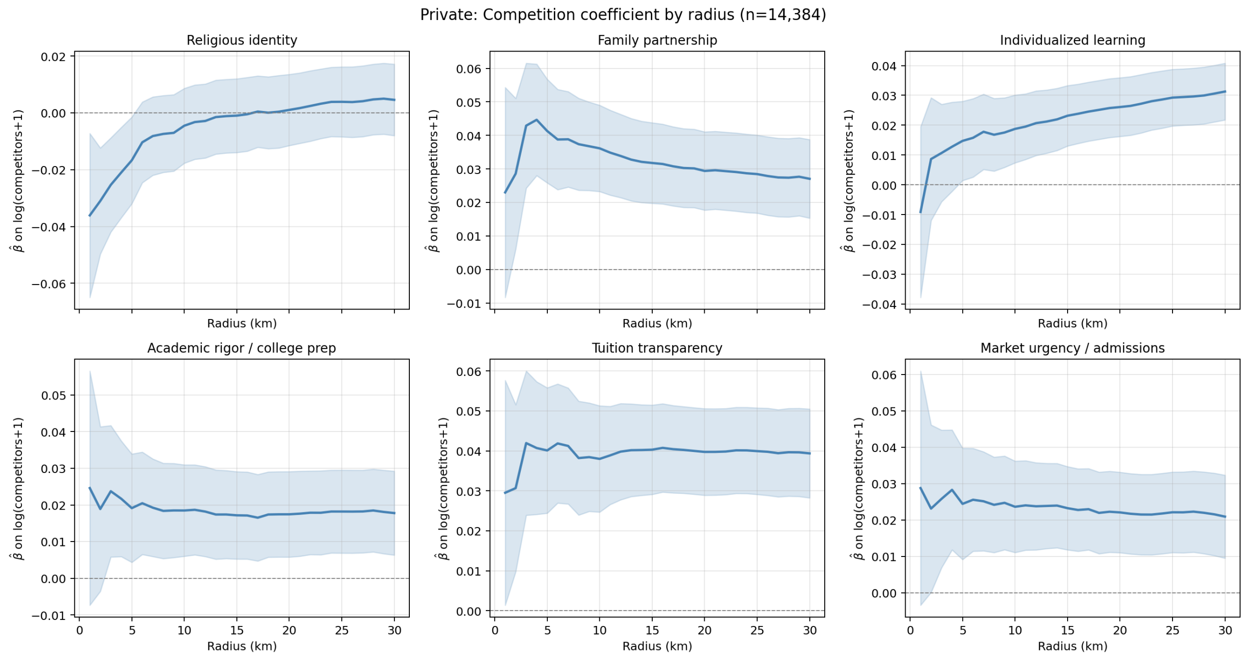
A measurement divergence to flag. The archived LLM-ontology version of this regression found family-partnership *prominence* declining monotonically with private competitor density. The keyword version reverses sign on that one topic family. The substantive issue is what each layer measures. The LLM atom scored prominence of the high-touch family-community claim, normalized at the page level. The keyword density score scales with overall content volume on parent-engagement topics, so it captures organizational machinery (PTA references, volunteer-coordination pages) more than the distinctive specialist claim. We retain the LLM finding in Appendix B and report the keyword-density divergence transparently in Appendix B.7. Both measurement layers agree on the central qualitative result: religious identity is the one dimension that does not respond to local market structure.

Table 11: Private-Sector Competition Regression: Keyword Topic Score on Log Competitor Count

Topic family	5 km	10 km	15 km	20 km
Religious identity	-0.0167**	-0.0045	-0.0010	+0.0010
	(0.0078)	(0.0067)	(0.0066)	(0.0064)
Family partnership	+0.0413***	+0.0361***	+0.0318***	+0.0294***
	(0.0079)	(0.0066)	(0.0061)	(0.0060)
Individualized learning	+0.0147**	+0.0188***	+0.0232***	+0.0261***
	(0.0068)	(0.0058)	(0.0052)	(0.0050)
Academic rigor / college prep	+0.0191**	+0.0185***	+0.0172***	+0.0174***
	(0.0075)	(0.0064)	(0.0061)	(0.0059)
Tuition transparency	+0.0401***	+0.0380***	+0.0403***	+0.0397***
	(0.0080)	(0.0068)	(0.0057)	(0.0055)
Market urgency / admissions	+0.0244***	+0.0236***	+0.0232***	+0.0221***
	(0.0078)	(0.0064)	(0.0058)	(0.0057)
n schools	14,384			

Notes: Each cell is the OLS coefficient on log(same-sector private competitors within radius + 1) from a separate regression with the named keyword topic score (z-scored within sector) as the dependent variable. Sample: $n = 14{,}384$ private schools with PSS coordinates. Competitor pool: 22,345-school PSS private universe. Controls: state, PSS affiliation, PSS typology fixed effects; log enrollment. HC1 standard errors in parentheses. Full 1–30 km radius curve in `table_competition_keyword_private_v4_20260519_all_radii.csv`. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 10: Private Competition Coefficient by Radius, 1–30 km



Notes: Six-panel coefficient plot. Each panel traces $\hat{\beta}$ on $\log(\text{competitors} + 1)$ for the named topic family across radii 1–30 km. Shaded band: 95% HC1 confidence interval. Religious identity sits on the zero line; the other five topics show positive monotone gradients.

Public-sector results and the cross-sector contrast

The public-sector regression on 66,678 schools delivers positive gradients on every topic family, with magnitudes 3× to 4× larger than the private-side coefficients at the same radii. Family partnership leads at $\hat{\beta} = +0.144$ (SE $= 0.003$, $p < 0.001$ at 10 km), followed by academic rigor (+0.113), market urgency (+0.078), tuition transparency (+0.067), individualized learning (+0.068), and religious identity (+0.040). Religious identity, which is flat on the private side, has a small but significant positive public-side gradient. The most plausible interpretation is that the public-side religious-identity signal is driven by Christian charter schools and religiously-affiliated public academies in dense urban markets rather than by mainstream district public schools.

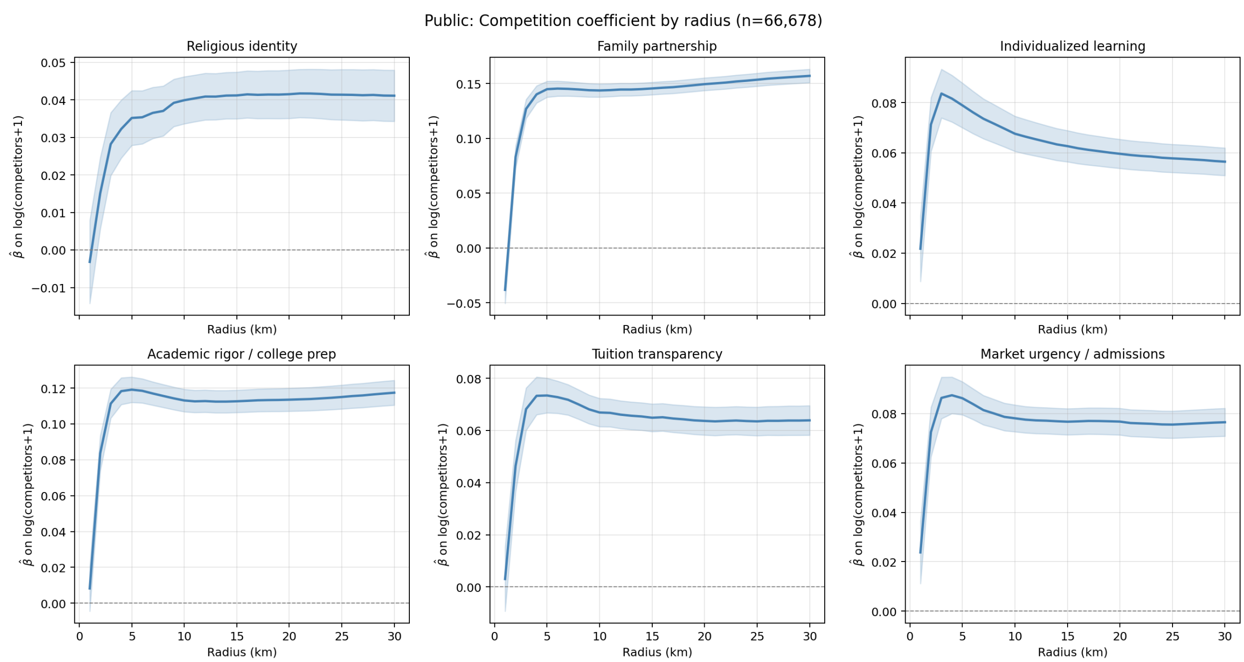
The cross-sector contrast in this keyword version is a magnitude difference rather than a sign reversal. Both sides show positive gradients; public schools in denser markets produce more content on every dimension than private schools in denser markets do. The archived LLM-prominence version found a sign reversal on family-partnership; the keyword version finds the same sign on both sides. Section 3.4 and Appendix B.7 discuss why the two measurement layers disagree on this particular margin.

Table 12: Public-Sector Competition Regression

Topic family	5 km	10 km	15 km	20 km
Religious identity	+0.0352***	+0.0399***	+0.0412***	+0.0415***
	(0.0037)	(0.0032)	(0.0032)	(0.0033)
Family partnership	+0.1450***	+0.1438***	+0.1456***	+0.1495***
	(0.0037)	(0.0031)	(0.0029)	(0.0029)
Individualized learning	+0.0789***	+0.0676***	+0.0627***	+0.0596***
	(0.0046)	(0.0036)	(0.0032)	(0.0030)
Academic rigor / college prep	+0.1192***	+0.1131***	+0.1127***	+0.1136***
	(0.0036)	(0.0032)	(0.0032)	(0.0033)
Tuition transparency	+0.0734***	+0.0669***	+0.0649***	+0.0637***
	(0.0034)	(0.0028)	(0.0027)	(0.0028)
Market urgency / admissions	+0.0862***	+0.0781***	+0.0767***	+0.0768***
	(0.0034)	(0.0028)	(0.0027)	(0.0028)
n schools		66,678		

Notes: Public-school analog of Table 11. Sample: n = 66,678 public schools with EDGE 2022–23 coordinates. Competitor pool: 102,157-school full EDGE public universe. Controls: state fixed effects, school-type fixed effects, charter dummy, log enrollment. HC1 standard errors in parentheses.

Figure 11: Public Competition Coefficient by Radius, 1–30 km



Notes: Public-school analog of Figure 10. All six topic families show positive monotone radius gradients, with magnitudes roughly 3–4× the corresponding private-side coefficients.

Reading the competition evidence

Two regularities survive the cross-sector pass. Religious identity is the single self-presentation dimension that local competition does not move on the private side: a market-independent signal tied to affiliation. Every other dimension scales with competitor density on both sides of the public-private split, with public magnitudes systematically larger. The substantive reading is not the specialist-vs.-generalist dichotomy that the LLM-prominence measure suggested in the archived draft. It is closer to a content-investment story: schools facing more local competitors produce more website content across most dimensions, and public schools (typically larger and operating district-shared infrastructure) produce more than private schools. Religious identity is the exception, and it is the informative one.

5.2 Community context

The cross-sectional community-context patterns from Section 4.5 return here as systematic regressions. We regress each topic-family score on county-level Republican two-party vote share (2020 election), adult bachelor's degree share (ACS 2014–18), log population density, log median household income, and the USDA Rural-Urban Continuum code, with state fixed effects throughout. The sample is the 79,261 crawled schools that merge cleanly to the 3,260-county national context frame (14,363 private and 64,887 public).

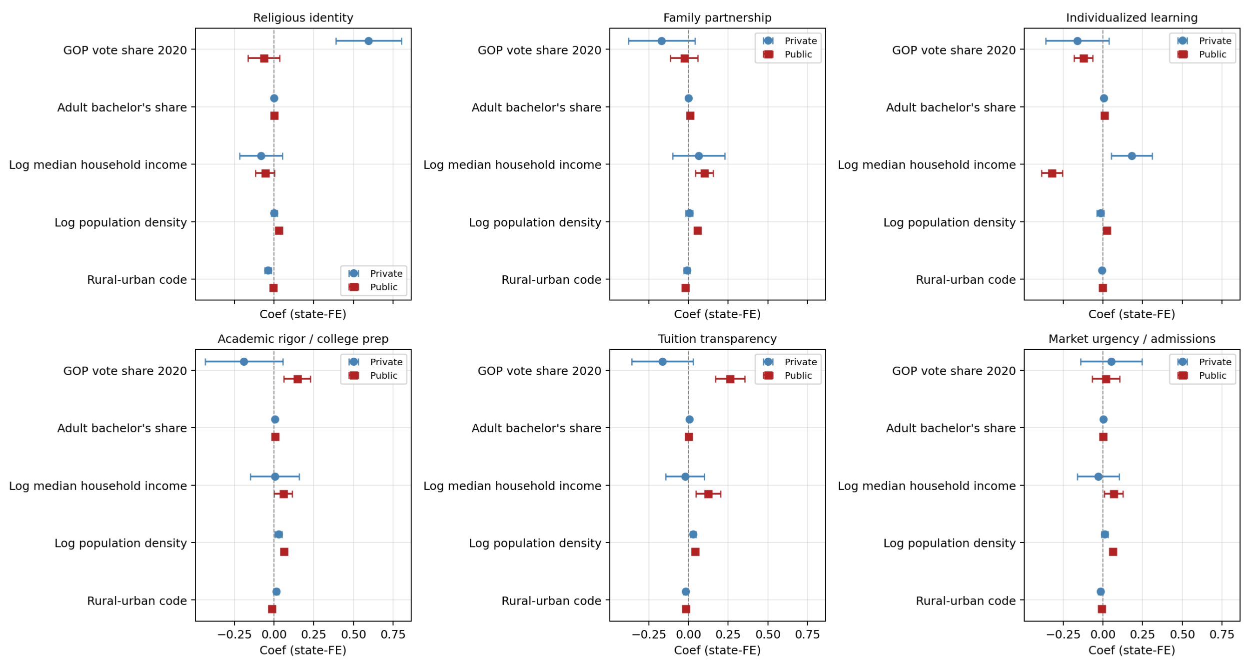
Table 13 and Figure 12 report the results. Two patterns dominate. On the private side, religious identity keyword density rises sharply with county Republican vote share ($\hat{\beta} = +0.60$, $p < 0.001$ at state-FE), confirming the descriptive correlation in Section 4.5 survives state demeaning. The signs of the other private-side coefficients on the GOP-vote share are negative for tuition transparency, individualized learning, and academic rigor (all $|\hat{\beta}| < 0.20$), and the bachelor's share moves all of those topics positively. On the public side, the same patterns hold but in attenuated form: individualized-learning keyword density is negative in Republican counties ($\hat{\beta} = -0.122$, $p < 0.001$); academic rigor and tuition transparency are positive (+0.15 and +0.26, both $p < 0.001$); religious identity is statistically flat (-0.07, *n.s.*), consistent with the institutional-identity reading.

Table 13: Community Context and Topic Prominence by Sector

	Religious identity	Family partnership	Individualized learning	Academic rigor / college prep	Tuition transparency	Market urgency / admissions
Panel: Private						
GOP vote share 2020	+0.596***	-0.168	-0.160	-0.189	-0.163*	+0.054
	(0.105)	(0.107)	(0.102)	(0.124)	(0.098)	(0.099)
Adult bachelor's share	-0.000	+0.000	+0.008***	+0.005***	+0.005***	+0.003*
	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)	(0.002)
Log median household income	-0.081	+0.065	+0.181***	+0.005	-0.021	-0.028
	(0.069)	(0.084)	(0.066)	(0.078)	(0.062)	(0.067)
Log population density	+0.002	+0.005	-0.016	+0.029**	+0.030***	+0.012
	(0.010)	(0.011)	(0.011)	(0.012)	(0.009)	(0.010)
Rural-urban code	-0.038***	-0.010	-0.004	+0.014*	-0.017**	-0.014**
	(0.010)	(0.010)	(0.006)	(0.008)	(0.007)	(0.007)
Panel: Public						
GOP vote share 2020	-0.065	-0.027	-0.122***	+0.145***	+0.261***	+0.018
	(0.051)	(0.044)	(0.030)	(0.042)	(0.047)	(0.044)
Adult bachelor's share	-0.001	+0.008***	+0.009***	+0.005***	+0.002*	-0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Log median household income	-0.056*	+0.100***	-0.321***	+0.057*	+0.125***	+0.067**
	(0.031)	(0.029)	(0.034)	(0.030)	(0.040)	(0.030)
Log population density	+0.031***	+0.057***	+0.023***	+0.063***	+0.042***	+0.061***
	(0.004)	(0.005)	(0.003)	(0.004)	(0.004)	(0.005)
Rural-urban code	-0.004	-0.020***	-0.002	-0.014***	-0.017***	-0.008***
	(0.003)	(0.003)	(0.002)	(0.002)	(0.003)	(0.003)

Notes: OLS regressions of topic score (z-scored within sector) on the listed county-level context variables with state fixed effects. Panels report private (top) and public (bottom) separately. HC1 standard errors in parentheses. Sample: 14,363 private + 64,887 public schools with all context variables non-missing. GOP vote share from 2020 county-level returns (`Poli2020Demos.csv`); adult bachelor's share, log median income, log population density, and rural–urban code from `CountyDemos.csv` (USDA / Census 2014–18). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 12: County-Level Context Correlates of Topic Prominence



Notes: Six-panel coefficient plot. Each panel shows the OLS coefficient on each county-context variable (with state fixed effects) for one focal topic. Blue: private. Red: public. Error bars: 95% HC1 CIs. Religious identity is the only topic where the private and public effects of GOP vote share have visibly different magnitudes.

5.3 Future determinants: the within-school panel

Several plausible determinants of website content can only be identified with within-school variation: external shocks (Trump administration on climate language; religious scandals on Catholic-school content; school shootings on safety language), local population changes (Latino or immigrant inflow, secularization), and competitor-entry or competitor-exit shocks. All require pre/post measurement on the same school's website. The Wayback Internet Archive panel under construction is designed to enable these tests in the next iteration of the paper. We mark them here as the natural continuation of the determinants agenda.

6 Consequences: Enrollment Growth and School Closure

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing. The recently-collected tuition$ continuous extraction, tract-level ACS demographics, and EDFacts achievement controls are wired into the appendix robustness (Sec. D.2); they will be folded into main-text regressions in the next iteration.]

Schools that show up differently on their websites attract different enrollment paths and survive at different rates. This section tests both predictions. We hedge causal language throughout because the cross-sectional website content reflects school identity, history, and prior market position; the within-school panel that would identify causal mechanisms is the topic of Section 7. The associations below are predictive.

6.1 Enrollment growth in the private sector

Specification

We estimate

$$\Delta \log E_{i,t+1} = \alpha + \sum_k \beta_k \, \text{TopicScore}_{ik} + \delta \, \mathbf{S}_i + \mathbf{X}'_{it}\gamma + \mu_t + \mu_s + \varepsilon_{it},$$

where $\Delta \log E_{i,t+1}$ is annualized log enrollment growth from PSS biennial waves 2013–14 through 2021–22 (half of biennial log growth, winsorized at the 1st and 99th percentile), TopicScore_{ik} is the standardized prominence-weighted keyword density on topic family k, and $\mathbf{S}_i$ collects the four deterministic transparency flags (tuition-amount-present, financial-aid-mention, accreditation-mention, test-scores-listed). Controls include year, state, PSS affiliation, and PSS typology fixed effects plus lagged log enrollment. Standard errors are school-clustered. The panel has 38,692 school-years across 12,437 private schools.

Headline results

The strongest growth predictor is whether a private school's website mentions financial aid: schools that mention it grow 1.77 percentage points faster per year (joint specification: $\hat{\beta} = +0.0177$, SE

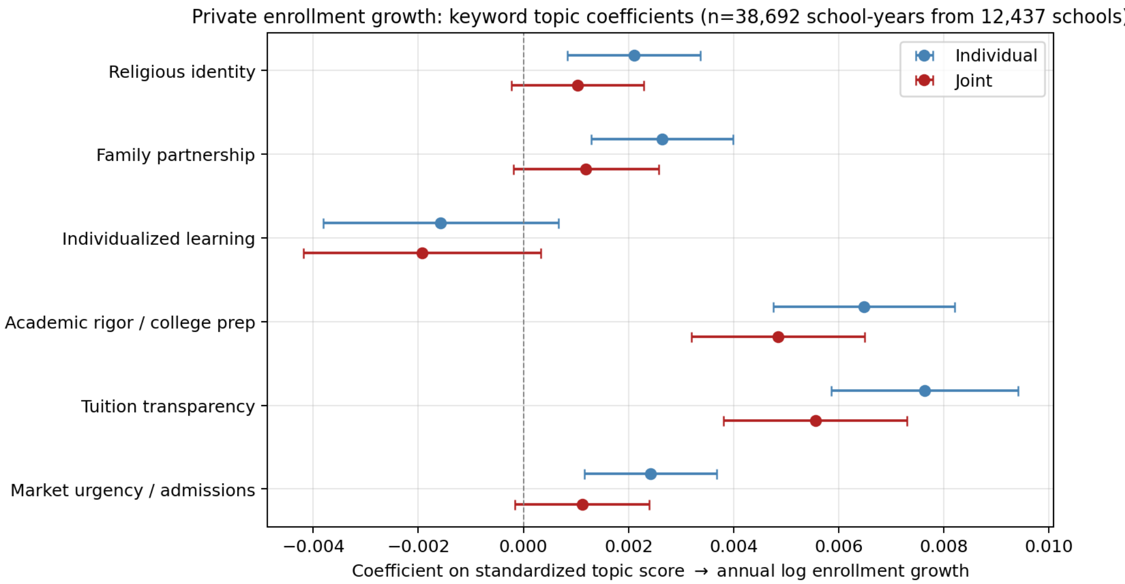
= 0.0020, $p < 0.001$). Tuition transparency keyword density is the strongest keyword-topic predictor at $\hat{\beta} = +0.0056$ (SE = 0.0009, $p < 0.001$): a one-standard-deviation increase shifts annualized growth by 0.56 percentage points. Academic rigor / college prep follows at $\hat{\beta} = +0.0048$ (SE = 0.0008, $p < 0.001$). Test-scores-listed adds +0.0085** and accreditation-mention adds +0.0034**. Religious identity, family partnership, individualized learning, and market urgency all have joint-spec coefficients within one standard error of zero, with 95 percent confidence intervals that exclude effects larger than 0.003 log points per standard deviation. These are precisely estimated zeros.

Table 14: Private-Sector Enrollment Growth: Keyword and Structural Predictors

	(1) Individual	(2) Joint
Religious identity	+0.0021***	+0.0010
	(0.0006)	(0.0006)
Family partnership	+0.0026***	+0.0012*
	(0.0007)	(0.0007)
Individualized learning	-0.0016	-0.0019*
	(0.0011)	(0.0012)
Academic rigor / college prep	+0.0065***	+0.0048***
	(0.0009)	(0.0008)
Tuition transparency	+0.0076***	+0.0056***
	(0.0009)	(0.0009)
Market urgency / admissions	+0.0024***	+0.0011*
	(0.0006)	(0.0007)
tuition_amount_present	—	+0.0031*
		(0.0017)
financial_aid_present	—	+0.0177***
		(0.0020)
accreditation_mentioned	—	+0.0034**
		(0.0016)
test_scores_listed	—	+0.0085***
		(0.0028)
n school-years	38,692	
n schools	12,437	
R^2 (joint)	0.0457	

Notes: Dependent variable: annualized log enrollment growth from PSS biennial waves 2013–14 through 2021–22, winsorized at 1st/99th percentile. Topic densities standardized to mean zero and unit variance within the private sector. Column (1): each predictor entered alone with full controls. Column (2): all predictors entered jointly. Controls in both: year, state, PSS affiliation, PSS typology fixed effects; lagged log enrollment. School-clustered standard errors in parentheses. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 13: Private-Sector Growth Coefficients



Notes: Coefficient plot from Table 14. Blue circles: individual specification. Red squares: joint specification. Error bars: 95 percent school-clustered confidence intervals. Tuition transparency, academic rigor, and the financial-aid structural flag sit clearly away from zero; the other topic families cluster around zero.

Translating the coefficient

A private school moving from the 25th to the 75th percentile of tuition-transparency keyword density (a shift of roughly 1.35 standard deviations) grows about 0.75 percentage points faster per year than an otherwise comparable school. The financial-aid mention implies a larger move: 1.77 percentage points per year, or 10 percent of the within-sample standard deviation of annual growth (which is roughly 0.17 log points in the private panel). For a private school with 100 students, the financial-aid-mention effect translates to about 1.8 additional students per year on average, compounding over time.

Measurement caveat: which growth predictor is the headline. The archived LLM-ontology version found individualized-learning prominence as the lead private-side growth predictor. The keyword version puts individualized learning in the precisely-estimated-zero column and replaces it with transparency and academic-rigor signals. The two measurement layers are measuring different things on this margin (Appendix B.7). The keyword version is the main-text analysis because it applies at $n = 12{,}437$ private schools rather than the LLM subsample's $n = 2{,}958$, and because the keyword dictionary is fully replicable.

6.2 Enrollment growth in the public sector and the sign reversal

Headline result

Academic-rigor keyword density flips sign between sectors. On the private side, $\hat{\beta} = +0.0048$ ($p < 0.001$): private schools whose websites foreground college-prep language grow faster. On the public side, the same measure carries $\hat{\beta} = -0.0013$ (SE = 0.0003, $p < 0.001$) in a panel of 251,792 school-years across 65,735 public schools. Public schools that lean into college-prep language are systematically schools losing enrollment, plausibly to charter and magnet alternatives.

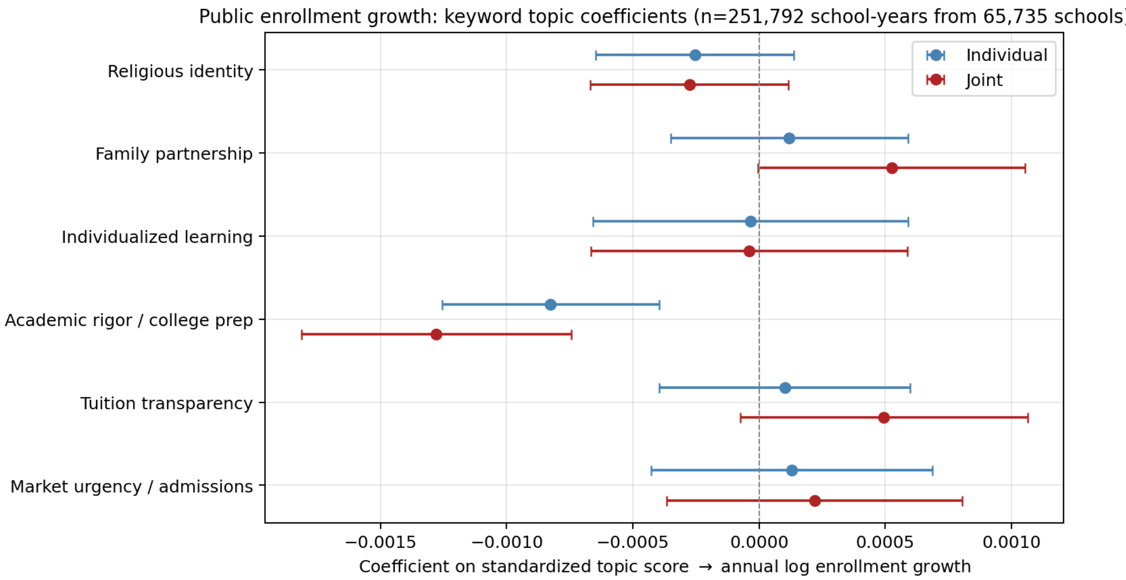
The remaining public-side coefficients are precisely estimated near zero. Religious identity, family partnership, individualized learning, tuition transparency, and market urgency all have joint-spec coefficients within one or two standard errors of zero. Among structural flags, only accreditation-mention shows a measurable positive coefficient. The financial-aid mention that dominates the private-side result is essentially zero on the public side — the signal is doing market-facing work in the private sector and bureaucratic-form work in the public sector.

Table 15: Public-Sector Enrollment Growth: Keyword and Structural Predictors

	(1) Individual	(2) Joint
Religious identity	-0.0003	-0.0003
	(0.0002)	(0.0002)
Family partnership	+0.0001	+0.0005*
	(0.0002)	(0.0003)
Individualized learning	-0.0000	-0.0000
	(0.0003)	(0.0003)
Academic rigor / college prep	-0.0008***	-0.0013***
	(0.0002)	(0.0003)
Tuition transparency	+0.0001	+0.0005*
	(0.0003)	(0.0003)
Market urgency / admissions	+0.0001	+0.0002
	(0.0003)	(0.0003)
tuition_amount_present	—	+0.0020
		(0.0015)
financial_aid_present	—	+0.0001
		(0.0006)
accreditation_mentioned	—	+0.0018**
		(0.0008)
test_scores_listed	—	-0.0000
		(0.0016)
n school-years	251,792	
n schools	65,735	
R^2 (joint)	0.0176	

Notes: Public-school analog of Table 14. Dependent variable: one-year log enrollment growth from the NCES CCD Membership file. Sample: $n = 251{,}792$ school-years across 65,735 schools, six waves stacked (CCD 2016–17 through 2022–23, omitting 2019–20). Controls: year, state, school-type fixed effects; charter dummy; lagged log enrollment. School-clustered standard errors in parentheses.

Figure 14: Public-Sector Growth Coefficients



Notes: Coefficient plot from Table 15. Academic rigor is the only topic with a clearly non-zero coefficient on the public side, and it carries the opposite sign from the private-side estimate in Figure 13.

Reading the cross-sector reversal

The same content claim — a website that foregrounds academic rigor and college-prep language — predicts opposite enrollment trajectories on opposite sides of the public-private split. Two readings are consistent with this. The first is sector-specific sorting: private schools that emphasize academic rigor attract families willing to pay tuition for that emphasis, so the language is associated with growing enrollment; public schools that emphasize academic rigor are disproportionately those facing selective competition from charter and magnet alternatives, so the language is associated with declining enrollment. The second is reverse causation: private schools that grow may invest more in their website (positive correlation by selection), while public schools that decline may invest more (negative correlation by selection). We cannot distinguish these without within-school variation. The cross-sector sign reversal does rule out a generic "a better website grows enrollment" story: the same content claim moves enrollment in opposite directions across sectors.

6.3 Population composition shapes who responds to which signal

The headline growth coefficients in Sections 6.1–6.2 are population-averaged. Coauthor review (PDF P23) raised an obvious extension: do these effects vary with the community a school serves? We interact each focal keyword topic z-score with two leading community-context variables (county GOP vote share, county adult bachelor's share, each z-scored within sector) inside the panel growth regressions, retaining year, state, and school-type fixed effects.

Table 16 reports the interaction coefficients. Three substantive results stand out. **Religious identity flips direction across communities on the private side**: the religious-identity growth gradient is more negative in Republican counties (interaction = -0.0019^{**}) and more positive in counties with higher bachelor's-share ($+0.0022^{***}$). On the surface this is the opposite of the most natural reading (religious schools in religious communities should attract more demand), but it makes economic sense given the descriptive cross-section: religious-identity prevalence is near-saturated in Republican counties already, so the marginal religious-identity signal is doing more differentiation work in lower-prevalence (higher-education) markets.

Individualized-learning content is more growth-protective in Republican counties (private interaction = $+0.0031^{**}$; public = $+0.0026^{***}$). On both sides of the public-private split, alternative pedagogy reads as a more salient differentiator where it is rarer. **Academic rigor and tuition transparency interactions are small on the private side and modestly positive on the public side**: public schools that signal college-prep or tuition language in Republican counties grow faster than public schools that signal the same in other counties (interactions $+0.0014^{***}$ and $+0.0007^{***}$), consistent with these signals being differentiating content in charter-competitive markets that concentrate in Republican states.

These interaction patterns confirm that the population-averaged coefficients in Tables 14 and 15 hide meaningful heterogeneity. They do not change the headline conclusions (transparency and rigor dominate on the private side; academic rigor flips sign on the public side), but they sharpen the substantive reading: each signal does different sorting work in different community types.

Table 16: Population × Topic Interaction Coefficients in Enrollment Growth Regressions

Sector	Context × topic interaction	Religious identity	Individualized learning	Academic rigor / college prep	Tuition transparency
Private	GOP vote share	-0.0019**	+0.0031**	+0.0001	-0.0007
		(0.0008)	(0.0013)	(0.0009)	(0.0008)
Private	Bachelor's share	+0.0022***	-0.0004	+0.0003	-0.0000
		(0.0008)	(0.0007)	(0.0007)	(0.0008)
Public	GOP vote share	+0.0004**	+0.0026***	+0.0014***	+0.0007***
		(0.0002)	(0.0008)	(0.0003)	(0.0002)
Public	Bachelor's share	-0.0000	-0.0003	-0.0005***	-0.0006***
		(0.0002)	(0.0006)	(0.0002)	(0.0002)

Notes: Interaction coefficients on (topic z-score × context z-score) in the panel growth regression with year, state, and school-type fixed effects plus lagged log enrollment, school-clustered standard errors. Private panel: $n = 38{,}627$ school-years across 12,418 schools (PSS biennial waves 2013–14 through 2021–22). Public panel: $n = 368{,}067$ school-years across 64,061 schools (CCD Membership 2016–17 through 2022–23, six waves). Context variables are county-level GOP two-party vote share (2020 election) and adult bachelor's share (ACS 2014–18). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

6.4 School closure: why the private panel is six states

A natural reaction to the six-state private closure panel below is to ask why we do not run this regression nationally. The answer is a timing constraint between features and outcome. Closure requires observing the school after the website measurement: a school that has already closed by crawl time no longer has a live website to extract features from. The 2026 features cannot predict PSS 2018, 2020, or 2022 closures, because every crawled private school in PSS 2020 is also in PSS 2022 (a school that closed before the crawl is gone from both the crawl and from subsequent PSS waves). To observe 2026 features and closure jointly we need a *post*-2022 enrollment-status source.

PSS 2023–24 will eventually provide one nationally but is not yet released by NCES. The 2024–25 state Department of Education enrollment rosters for California, New York, Pennsylvania, Illinois, Ohio, and Wisconsin are the only post-PSS-2022 enrollment sources we obtained at the school level; FL/VA/GA/IL-AG and the Cognia database were anti-bot-blocked or yielded zero events (Appendix D.4).

A complementary design uses *earlier* features and earlier closures across both sectors. Appendix D.3 reports a national all-school regression that extracts 2017–18 Wayback Machine homepage features for 25,780 schools (6,076 private + 19,704 public) and defines closure as absence from the corresponding 2021–22 roster (PSS for private, CCD Membership for public) — a 1,965-event national panel covering all 50 states and both sectors. Pooled headline: one-standard-deviation increases in academic-rigor / college-prep keyword density and family-partnership keyword density are each associated with roughly 15–19 percent lower closure odds (OR $\approx$ 0.81 and 0.85, both $p < 0.001$). Sector heterogeneity shows the academic-rigor protection concentrates in public schools and that religious-themed public schools are particularly slow to close.

For the public sector the data constraint does not bind: the National Center for Education Statistics publishes the Common Core of Data Membership file annually, and the 2022–23 wave gives us a post-crawl enrollment-status observation for every public school in the country. The national public closure panel below uses seven CCD waves and contains 9,344 closure events across 109,204 schools nationally — the largest closure-with-website-features panel in this paper.

Specification

We estimate a binary logit

$$\Pr(\text{Closure}_{i,t+1} = 1) = \Lambda\left(\alpha + \sum_k \beta_k \,\text{TopicScore}_{ik} + \delta\, \mathbf{S}_i + \mathbf{X}'_{it}\gamma + \mu_s + \varepsilon_{it}\right),$$

where closure means the school appears in the state Department of Education roster in year t but is absent in year $t+1$, with the absence persisting. The six-state panel (CA, NY, PA, IL, OH, WI) contains 1,553 matched private schools and 70 closure events. Affiliation breakdown: Nonsectarian (665 schools, 29 events), Other religious (609, 30), Catholic (279, 11). Controls: state fixed effects, PSS affiliation dummies, log enrollment.

Headline results

The same two predictors that dominate the private-side growth regression return as protective predictors of survival. Academic rigor / college prep carries a closure log-odds coefficient of -0.42 (SE = 0.25, $p = 0.09$; odds ratio $\approx$ 0.66): a one-SD increase in college-prep keyword density is associated with school closure at roughly two-thirds the odds of otherwise comparable schools. Financial-aid mention carries -0.52 (SE = 0.32, $p = 0.10$; odds ratio ≈ 0.59): schools that mention financial aid close at roughly 59 percent the odds. The other coefficients are noisy at 70 events.

Pseudo-$R^2 = 0.10$.

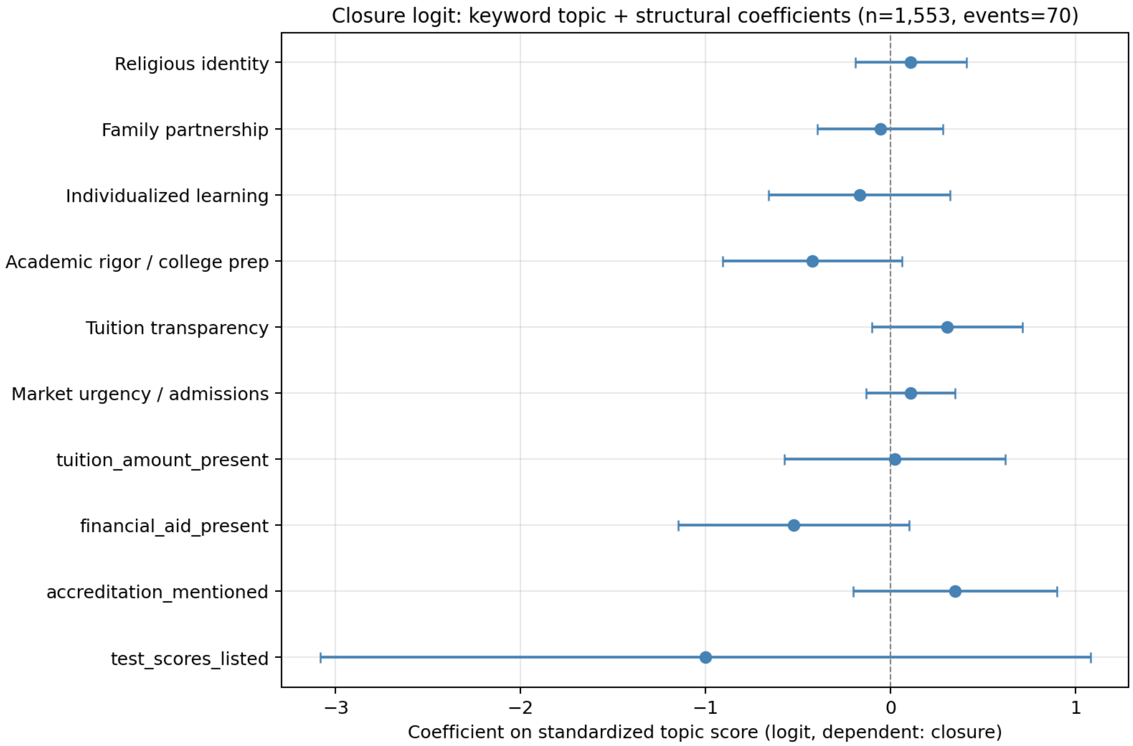
The two protective signals are the same two that dominate the private-side growth regression. Schools that disclose price-and-aid information and that foreground quality language are schools that both attract families and survive in the panel.

Table 17: School Closure: Pooled Keyword and Structural Predictors

	$\hat{\beta}$	Odds ratio
Religious identity	+0.1094	1.116
	(0.1537)	
Family partnership	-0.0560	0.946
	(0.1728)	
Individualized learning	-0.1681	0.845
	(0.2499)	
Academic rigor / college prep	-0.4237*	0.655
	(0.2480)	
Tuition transparency	+0.3051	1.357
	(0.2076)	
Market urgency / admissions	+0.1066	1.112
	(0.1229)	
tuition_amount_present	+0.0240	1.024
	(0.3048)	
financial_aid_present	-0.5239	0.592
	(0.3188)	
accreditation_mentioned	+0.3489	1.418
	(0.2814)	
test_scores_listed	-1.0018	0.367
	(1.0616)	
n schools	1,553	
Closure events	70	

Notes: Binary logit. Dependent variable: school appears in DOE roster in year t but absent in $t+1$, persisting. Six-state panel (CA, NY, PA, IL, OH, WI) with 2024–25 endpoints for CA, NY, PA, IL and 2023–24 for OH, WI. Sample: $n = 1{,}553$ schools, 70 events. Topic densities z-scored within the private sector. Controls: state fixed effects, PSS affiliation fixed effects, log enrollment. HC1 standard errors in parentheses; pseudo-$R^2 = 0.10$.

Figure 15: Closure Logit Coefficients



Notes: Coefficient plot from Table 17. Academic rigor and financial-aid-mention sit clearly on the protective side; other predictors cluster near zero.

The within-Nonsectarian finding does not replicate at keyword presence

The archived LLM-ontology draft reported a within-Nonsectarian protective effect of religious-and-moral content presence (odds ratio ≈ 0.37, $p < 0.05$). The keyword analogue of that specification does not replicate. Table 18 reports the comparison. The within-Nonsectarian coefficient is $+0.30$ (SE $= 0.76$, *n.s.*; odds ratio ≈ 1.34). Catholic and Other-religious cells are unidentified at the presence margin because prevalence approaches 100 percent in those groups.

The keyword null reflects a measurement issue, not an absence of pattern. The keyword "any religious-and-moral content" indicator was built as the max of `kw_religious_present`, `kw_character_present`, and `kw_civic_present`. The character dictionary includes broad words — "responsibility," "service," "honor," "integrity" — that appear on 92 percent of Nonsectarian websites and 95–98 percent of religious-school websites. With that prevalence the binary presence indicator carries essentially no within-Nonsectarian variation, and the LLM atom's 0–3 prominence scale could detect prominence that the keyword-presence flag cannot. A cleaner keyword analogue would restrict to `kw_religious_present` alone or use a continuous-density threshold. We retain the LLM-version finding in Appendix B and treat the keyword version of this specific test as inconclusive.

Table 18: Closure by Affiliation: Religious-and-Moral Presence (Keyword Version)

	(1) Pooled binary	(2) By affiliation
Any religious-and-moral content	+0.4652	—
	(0.6154)	
R&M presence × Catholic	—	-1.3283
(n=279, prev=0.97)		(1.1587), OR = 0.265
R&M presence × Nonsectarian	—	+0.2952
(n=665, prev=0.92)		(0.7618), OR = 1.343
R&M presence × Other religious	—	+20.2712
(n=609, prev=0.95)		(18304.2525), OR = 636306862.050
n schools	1,553	
Closure events	70	

Notes: Binary logit with the keyword "any religious-and-moral content" indicator (max of `kw_religious_present`, `kw_character_present`, `kw_civic_present`) interacted with PSS affiliation. Column 1: pooled binary. Column 2: affiliation-specific binary indicators. Catholic and Other-religious cells unidentified at the presence margin (prevalence $\geq$ 95 percent). The within-Nonsectarian effect is indistinguishable from zero in the keyword version. The corresponding LLM-ontology result (odds ratio ≈ 0.37, $p < 0.05$) is in Appendix B.

6.5 School closure: public sector at national scale

The 70-event private-side panel is event-thin. The public sector is the natural place to ask whether the same content predictors generalize.

Construction of the national public-school closure panel

We track each school's presence across seven NCES Common Core of Data Membership waves (2016–17 through 2022–23). A school is coded as *closed in panel* if it appears in some wave t but disappears in all subsequent waves through 2022–23. The panel contains 109,204 unique public schools and 9,344 closure events (8.6 percent of schools closed over the seven-year window). Cross-validation against the CCD 2022–23 directory `SY_STATUS_TEXT` field, which independently flags 878 schools as *Closed*, confirms that the panel definition captures the same closures the directory does. The closure year is the year after the school's last-observed membership wave.

Merging the closure panel with the v3 features file leaves 66,611 schools and 320 closure events for the regression (closures concentrate in schools that never made it into the 2022–23 directory after the crawl, so the post-merge event count is much smaller than the panel total).

Headline results

Three coefficients reach $p < 0.05$ on the public side (Table 19, Figure 16). Religious-identity keyword density is the most protective predictor ($\hat{\beta} = -0.96$, SE = 0.34; odds ratio ≈ 0.38): public schools with religious-identity keywords on their websites close at roughly 38 percent the odds of otherwise comparable schools. The most plausible reading is sector-compositional: religiously-affiliated charters and religious-themed public academies, which carry the keyword, are also schools embedded in denominational networks that buffer them against closure. Tuition-transparency keyword density and the financial-aid-mention structural flag carry the same direction as on the private side (OR = 0.70 and OR = 0.54 respectively, both $p < 0.05$). The other topic coefficients and structural flags are indistinguishable from zero.

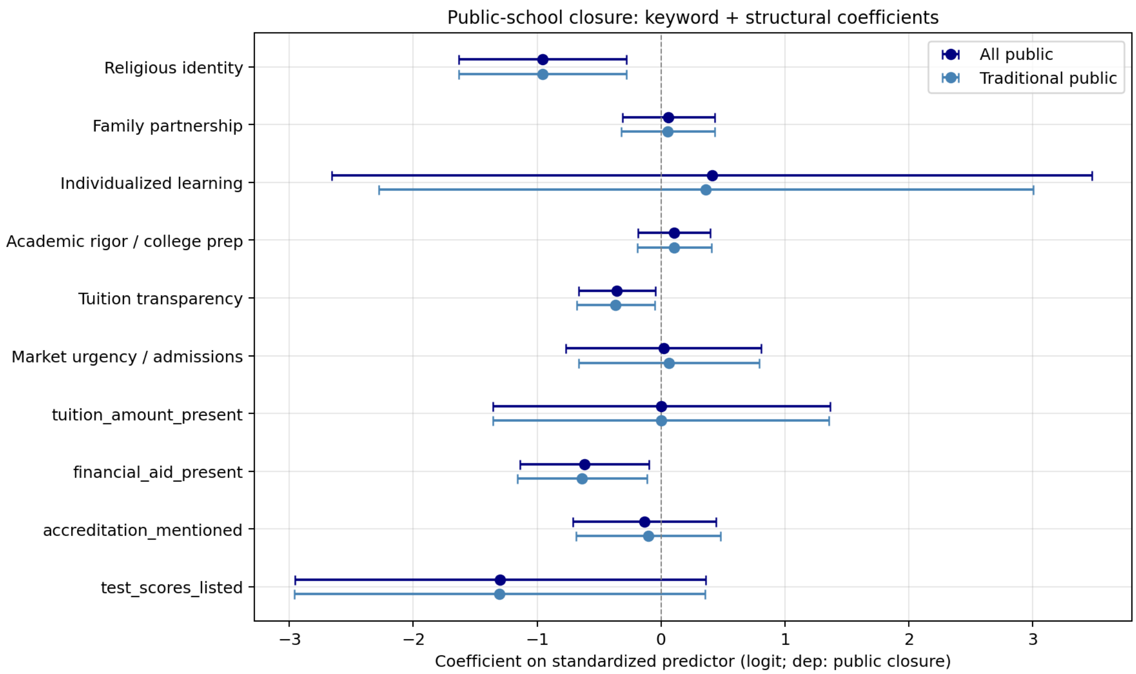
Charter schools register zero closures in the panel and are dropped from the segment regression. Charters that cease operations typically rename, transition, or reorganise under a new NCESSCH rather than disappear from the membership rolls, so the closure-definition we use does not capture the charter equivalent of school termination.

Table 19: Public-School Closure: National Panel

	All public	Traditional public
Religious identity	-0.956***	-0.958***
	(0.344)	(0.345)
Family partnership	+0.060	+0.054
	(0.191)	(0.193)
Individualized learning	+0.412	+0.362
	(1.565)	(1.348)
Academic rigor / college prep	+0.106	+0.106
	(0.149)	(0.153)
Tuition transparency	-0.356**	-0.366**
	(0.158)	(0.161)
Market urgency / admissions	+0.020	+0.063
	(0.402)	(0.371)
tuition_amount_present	+0.003	+0.000
	(0.694)	(0.692)
financial_aid_present	-0.618**	-0.637**
	(0.265)	(0.267)
accreditation_mentioned	-0.134	-0.103
	(0.294)	(0.297)
test_scores_listed	-1.298	-1.303
	(0.846)	(0.845)
n schools	66,611	61,842
Closure events	320	320
Pseudo R^2	n/a	n/a

Notes: Binary logit on the national public-school closure panel. Dependent variable: school appears in at least one CCD Membership wave (2016–17 through 2022–23) but disappears in all later waves. Sample: 66,611 public schools merged with the v3 features file; 320 closure events. Topic densities z-scored within the public sector. Controls: state fixed effects, log enrollment. HC1 standard errors in parentheses; ridge penalty 10^{-3} for stability. Charter segment excluded (0 closures in panel). $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 16: Public-School Closure Logit Coefficients



Notes: Coefficient plot from Table 19. Religious-identity keyword density, tuition-transparency keyword density, and financial-aid-mention flag are the three predictors with confidence intervals clearly excluding zero. The remaining predictors cluster around zero.

[Suggestive — revisit] A complementary post-crawl design has recently become feasible with the release of CCD Membership 2024–25 (NCES, July 2025). Pairing the May 2026 features with the 2022–23 → 2024–25 closure window yields a national public panel of 64,458 schools with 1,357 closures. Headline topic coefficients in that panel *flip sign* relative to the seven-wave panel above: academic-rigor / college-prep density +0.062 ($p = 0.010$), tuition-transparency +0.055 ($p = 0.016$), and market-urgency +0.072 ($p = 0.004$) are now associated with *higher* closure odds. School-level EDFacts achievement data (2016–17 / 2017–18 average percent proficient) suggests this is a "distress-marketing" pattern: schools that closed by 2024–25 had lower prior achievement (41.5% vs 48.9% proficient among survivors) but higher 2026 college-prep keyword density (68.0 vs 53.7). The correlation between any keyword score and prior achievement is near zero ($r \leq 0.08$), so the 2026 web content does not reflect underlying quality; closing schools appear to be ramping up college-prep and admissions language while their measured outcomes decline. This is suggestive only — the same schools may also be dropped from CCD for administrative reasons unrelated to economic closure — and the regression sign-flip warrants its own investigation before being elevated to a headline finding. To revisit in the next iteration with the multi-period within-school panel from the Wayback + Common Crawl pipelines.

The cross-sector closure pattern is consistent: tuition-transparency and financial-aid signals are protective on both sides of the public-private split. Religious-identity is protective on the public side (where it captures denominationally networked schools) and noisy on the private side (where it is too prevalent within Catholic and Other-religious affiliations to identify a within-group effect at the binary margin).

6.6 Reading the consequences evidence

The keyword-level evidence delivers a consistent two-finding story on the private side: tuition-transparency content (the financial-aid mention, plus tuition-transparency keyword density) and academic-rigor content together predict both faster enrollment growth and lower closure odds. The two signals do not just predict different margins of demand; they predict the same direction of demand response across two outcomes that share the underlying mechanism (families attracted to schools that offer clear price information and clear quality claims).

The public-side evidence delivers one substantively important result — the academic-rigor sign reversal — and otherwise precisely estimated zeros. Public-school website content does not move public-school enrollment in the same way that private-school content moves private enrollment, with the single exception that selective-language content marks public schools that are losing students. Whether this is sector-specific sorting or reverse causation cannot be distinguished without the within-school panel.

The archived LLM-ontology version of the consequences section identified slightly different headline predictors — individualized learning for private growth, religious-and-moral identity for within-Nonsectarian closure protection. Both findings are preserved in Appendix B on the 2,958-school LLM subsample. Both measurement layers agree on the broad qualitative pattern: website content predicts both growth and survival, with the strongest signals coming from content that helps families resolve the information friction (price, aid, academic claim) that the U.S. K–12 market does not regulate.

7 Conclusion

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing. The recently-collected tuition$ continuous extraction, tract-level ACS demographics, and EDFacts achievement controls are wired into the appendix robustness (Sec. D.2); they will be folded into main-text regressions in the next iteration.]

The U.S. K–12 market enrolls more than 50 million students, but no regulator requires schools to disclose what families need to choose them. We construct a national, cross-sector, fully replicable measurement of what 81,109 public, private, and charter schools tell families on their websites, and we use that measurement to ask three questions. What do schools say? What drives them to say it? What does what they say predict?

The descriptive cross-section delivers a clean sector contrast. Private schools dominate every market-facing transparency signal: tuition disclosure, financial-aid mentions, accreditation language, mission-page presence. Charter schools sit between private and traditional public on most dimensions and lead both on market-urgency language and on individualized-learning vocabulary. Religious-identity content remains a private-sector phenomenon. The cross-sector content layer carries information that researcher-assigned categories alone do not.

The determinants evidence isolates the single dimension that does not respond to local market structure. Religious-identity keyword density is flat across private-school competitor density at every radius from 1 to 30 km. Every other focal topic — family partnership, tuition transparency, market urgency, individualized learning, academic rigor — has a positive coefficient on log competitor count, with public-side magnitudes $3\times$ to $4\times$ the private-side magnitudes. Religious identity is the exception, and identity economics provides the natural mechanism: a school's religious self-presentation responds to affiliation, not to block-level alternatives.

The consequences evidence converges on a two-signal headline for the private sector. Tuition-transparency keyword density and the financial-aid-mention flag together predict both faster enrollment growth (the financial-aid mention alone implies $+1.77$ percentage points per year) and lower closure odds (odds ratio ≈ 0.59). Academic-rigor keyword density predicts private growth in the same direction and flips sign in the public sector (-0.0013, $p < 0.001$), where college-prep language marks schools losing students to charter and magnet alternatives. The same content claim moves enrollment in opposite directions across sectors — a result that rules out a generic "better website grows enrollment" story and points to sector-specific sorting on observable content.

For school-choice policy, the implication is that information families need already exists in observable form on school websites. The U.S. K–12 market has no mandated price disclosure, but every private school that chooses to disclose financial-aid information signals roughly 1.8 percentage points of additional annual growth and 40 percent lower closure odds. The measurement infrastructure we build can be applied at near-zero marginal cost to any future corpus, including historical Internet Archive snapshots. Families with the narrowest social networks and the least time to research stand to benefit most from standardized website measurement: such families are the least equipped to extract signal from unstructured marketing pages and the most exposed to closure risk on the schools they currently attend.

For school operators, the message is narrower but more actionable. Transparency about price and financial aid does not appear to drive families away; it appears to attract them, conditional on the school being able to deliver the academic-rigor signal that the same families respond to. The operational lesson is the conjunction: clear price disclosure plus clear quality language, on a website that loads cleanly and includes a working admissions page.

Two extensions sit at the top of the data agenda. The first is the within-school Internet Archive panel. We have already retrieved Wayback snapshots for 4,854 schools, including 7,930 with a snapshot in calendar years 2017–2018. *This 7,930-school subset is the current Internet Archive coverage; expanding it through additional Wayback CDX queries and a Common Crawl AWS Athena pass would raise the panel several-fold relative to the current iteration — see the within-school analysis in Appendix C for the historical-coverage roadmap.* A pilot fetch of the 2017–2018 homepages for the 7,930 subset schools has now run (Appendix C); applying the deterministic measurement layer to 2018 and 2026 snapshots of the same school yields a within-school change panel that controls for fixed school identity. Instead of asking whether high-tuition-transparency schools survive, the panel allows asking whether *adding* a financial-aid mention between 2018 and 2026 predicts subsequent enrollment growth or closure — the cleanest identification fix this paper does not yet make. The pilot panel of 7,930 schools is a meaningful start but is a small subset of the national universe; subsequent iterations will expand Wayback coverage and integrate the Common Crawl S3 columnar index for additional historical capture.

The second extension is the national closure-panel expansion that this paper has already begun on the public side: 109,204 public schools across seven CCD waves produces 9,344 closures (Section 6.5). Bringing the private side to the same scale requires either adding state Department of Education enrollment files for the remaining 44 states or exploiting PSS's own school-status flags — both feasible data extensions that would move the private closure panel from 70 events to the several hundred required for atom-level intensity precision.

The deterministic measurement layer is a public good. The keyword dictionary, extraction script, and feature file can be released as a replication artifact at submission. The same dictionary applied to a future U.S. Wayback corpus, to a Canadian or U.K. K–12 corpus, or to a college-level corpus recovers the same primitive features at the same cost. The version of text-as-economic-data this paper demonstrates scales without proprietary inference, and the next researcher who wants to ask similar questions can do so without our help.

Appendix

Contents

A Measurement Notes and Coverage Diagnostics

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing.]

This appendix collects measurement and coverage details that support the main-text data section but would interrupt the narrative if placed inline.

A.1 Coverage by state and sector

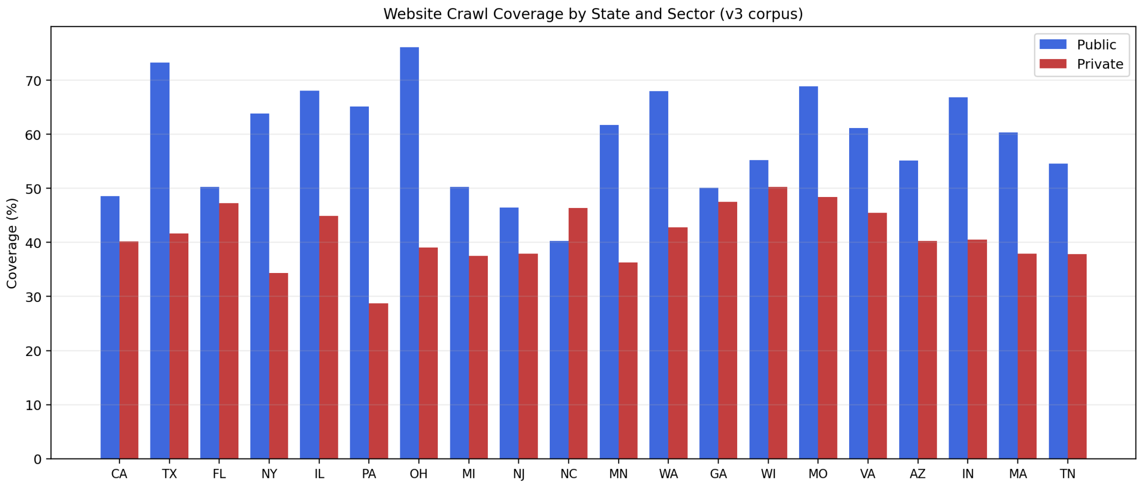
Table 1 in the main text reports overall coverage of 93.3 percent for active schools. The breakdown by state and sector exposes meaningful heterogeneity. Coverage is highest in states with well-maintained NCES CCD directories (most of the Midwest and Mid-Atlantic) and lowest where private-school official spines are thin or where district-integrated schools are not separately webbed (West Virginia is the extreme case; rural West Virginia public schools are typically housed on district-level pages with no separate school URL).

Table 20: URL Coverage by State and Sector

State	Sector	Seeded	Crawled	Coverage	Pages	Avg pages	Cap-hit share
HI	public	297	63	21.2%	347	5.51	82.5%
NC	public	2812	1148	40.8%	4982	4.34	45.6%
VT	public	514	234	45.5%	1175	5.02	70.5%
DC	public	250	114	45.6%	539	4.73	67.5%
NJ	public	2630	1253	47.6%	6561	5.24	69.4%
DE	public	240	119	49.6%	515	4.33	51.3%
AL	public	1559	777	49.8%	3952	5.09	69.0%
CT	public	1173	596	50.8%	3077	5.16	70.6%
CA	public	10957	5712	52.1%	30390	5.32	71.6%
MI	public	3841	2001	52.1%	10678	5.34	73.1%
MS	public	1095	581	53.1%	3248	5.59	84.3%
WI	public	2377	1319	55.5%	7222	5.48	79.5%
FL	public	4320	2406	55.7%	11263	4.68	59.8%
TN	public	1974	1104	55.9%	5185	4.70	54.3%
AZ	public	2555	1477	57.8%	7521	5.09	61.2%

Notes: Lowest-coverage state-sector cells, sorted ascending by URL coverage. Cells with fewer than 100 seeded schools are excluded.

Figure 17: URL Coverage by State and Sector



Notes: Public coverage (upper bars) vs. private coverage (lower bars) by state. The public short-tail concentrates in West Virginia and a few small-state CCD-directory gaps; the private short-tail is more uniformly distributed.

The unseeded tail is not a pipeline failure. Three pieces of evidence support that reading. First, every active public school without a seed URL also has no URL in NCES's own CCD directory `WEBSITE` field — the federal statistical agency itself has no record for these schools. Second, unseeded active private schools enroll a median of 39 students against an active-private median of 82, with 56 percent enrolling 50 students or fewer — school sizes typical of home-based or single-family programs. Third, a stratified spot-check on 609 unseeded schools found 94 percent of large unseeded public schools (500+ enrollment) via independent web search — pipeline misses now being recovered — but essentially zero among small private schools (under 100 enrollment) and zero in West Virginia. The 6.7 percent unseeded tail among active schools is mostly genuinely unwebbed, not missed.

A.2 Charter, traditional public, and private sense-check

The main text reports descriptive headlines pooling traditional public and charter schools. Many readers will want to see the breakdown. Table 21 reports it.

Table 21: Private, Traditional Public, and Charter School Self-Presentation

	Private	Public traditional	Public charter
N schools	14,384	61,956	4,769
Website investment (per-school cap = 50 unique pages)			
Raw pages retrieved (mean)	23.8	23.7	22.1
Unique pages after within-school dedup (mean)	23.0	22.7	21.1
Total word count, deduplicated (mean)	33,140.1	34,450.3	29,711.9
Keyword topic prevalence (% with any hit)			
Religious identity	70.0%	23.8%	21.3%
Family partnership	64.0%	64.5%	67.5%
Individualized learning	36.2%	14.8%	35.8%
Academic rigor / college prep	78.1%	76.8%	81.3%
Tuition transparency	82.8%	49.8%	68.4%
Market urgency / admissions	40.7%	29.4%	65.4%
STEM	73.1%	67.1%	72.5%
Arts / athletics	78.6%	76.6%	77.1%
DEI / inclusion	49.2%	43.3%	54.5%
Sustainability	56.8%	50.3%	55.2%
SEL / wellness	58.3%	63.0%	68.7%
Character / civic	91.8%	85.0%	88.3%
Structural/transparency flags (% present)			
Tuition amount (regex $)	38.8%	3.4%	4.1%
Financial aid mentioned	60.6%	30.4%	34.8%
Accreditation mentioned	47.2%	14.7%	24.8%
Test scores listed	4.4%	3.0%	3.9%
Page-type presence (% with ≥ 1 page)			
Mission / about page	79.1%	58.8%	68.6%
Admissions page	60.0%	32.8%	51.3%
Tuition page	35.8%	2.5%	3.0%
Athletics page	21.3%	16.2%	12.8%
Arts page	16.1%	8.5%	9.0%
News / blog	32.3%	24.2%	30.6%
Policy / handbook	21.0%	28.1%	26.7%

Notes: Unit of observation: school. Three segments: private (PSS, $n = 14{,}384$), traditional public (CCD non-charter, $n = 61{,}956$), and charter (CCD charter, $n = 4{,}769$). Website-investment rows: mean values. Other rows: percentage of crawled schools in segment with the listed signal present. Source: `features_national_20260517_v3.csv` merged with the national school master. Charter classification: CCD `is_charter` field.

Three regularities surface from the three-way split. First, charter schools are operationally similar to traditional public schools (similar word counts, similar mission-page presence) but resemble private schools on admissions-pathway content (51 percent admissions-page presence vs. 33 percent traditional public). Second, charter schools lead both other segments on market-urgency language: 65 percent of charter websites use “apply now,” “limited seats,” or “deadline” vocabulary, against 41 percent private and 29 percent traditional public. Third, charters match private schools on individualized-learning vocabulary (36 percent vs. 36 percent private; only 15 percent traditional public), consistent with the charter sector's institutional position as alternative-pedagogy public schools.

The pricing dimensions are still essentially private-sector phenomena. Tuition disclosure (regex match on dollar values) appears on 38.8 percent of private websites, 3.4 percent of traditional public, and 4.1 percent of charter. Financial-aid mentions appear on 61 percent of private, 30 percent of traditional public, and 35 percent of charter websites. The charter financial-aid share above the traditional-public share reflects charter schools that serve subsidized after-care, scholarship, or busing programs for low-income families.

A.3 Keyword dictionary specification

The full keyword dictionary is in `scripts/84ca_keyword_structural_extractor.py`, with one regex pattern per topic family. We summarize the 22 topic families and provide a representative sample of keywords for each in Table 22; the script itself is the canonical specification.

Table 22: Keyword Dictionary: 22 Topic Families and Sample Keywords

Topic family	Sample keywords
`kw_religious`	faith, prayer, god, scripture, chapel, worship, biblical, christ, jesus, lord
`kw_denominational`	catholic, christian, jewish, islamic, muslim, lutheran, baptist, methodist, episcopal, presbyterian
`kw_character`	integrity, responsibility, grit, character, service, honor, virtue, moral, ethics
`kw_civic`	patriot, civic, democracy, citizenship, flag, american values, pledge of allegiance
`kw_college_prep`	ap, advanced placement, IB, honors, dual enrollment, college acceptance, SAT, ACT
`kw_curriculum_philosophy`	montessori, waldorf, classical education, project-based, inquiry-based, socratic
`kw_stem`	stem, coding, robotics, maker space, engineering, 3d printing
`kw_arts`	visual arts, performing arts, theater, music, dance, band, choir, orchestra
`kw_technology`	1:1, chromebook, canvas, google classroom, technology integration
`kw_gifted`	gifted, GATE, accelerated, enrichment program, advanced learner
`kw_learning_support`	IEP, learning differences, dyslexia, special education, 504 plan
`kw_athletics`	athletics, varsity, interscholastic, championship, sports team
`kw_parent_involvement`	parent association, PTA, PTO, volunteer, family engagement
`kw_community_service`	community service, service learning, outreach, volunteer hours
`kw_alumni`	alumni, alumnae, graduates, alumni network, class of
`kw_dei`	diversity, equity, inclusion, belonging, multicultural
`kw_sustainability`	sustainability, sustainable, green, environmental, climate
`kw_sel`	SEL, social-emotional, mindfulness, wellness, whole child
`kw_urgency`	limited seats, apply now, enrollment open, waitlist, deadline
`kw_differentiation`	unique approach, alternative school, unlike traditional
`kw_social_proof`	award-winning, blue ribbon, ranked, US News, accredited
`kw_outcome_claims`	100% college, placement rate, alumni success, college bound
`kw_tuition_transparency`	tuition, fees, cost of attendance, financial aid, scholarship

Notes: Each entry is a regex pattern in `scripts/84ca_keyword_structural_extractor.py`. Multi-word phrases match as substrings; single words use word boundaries. Prominence weighting: homepage hits $\times 3$, heading hits $\times 2$, body hits $\times 1$, deeper-page hits $\times 0.5$. Per-school output: presence flag, raw count, prominence-weighted score.

A.4 Page-type detection rules

Each crawled page is classified into at most one page type from twelve categories: admissions, tuition, academics, mission/about, athletics, arts, staff, news/blog, calendar, virtual tour, library, policy/handbook. Classification uses a fall-through rule on three signals in order: URL path keyword (e.g., `/admissions/`), page title keyword, and first-heading keyword. The school-level binary flag for page type T is 1 if any crawled page on the school's site classifies into T. Detection rules and the full keyword list per page-type are in `scripts/84ca_keyword_structural_extractor.py`.

A.5 Structural feature definitions

`n_pages_crawled_raw`: total crawled pages including duplicates and boilerplate. `n_unique_pages`: distinct page hashes after deduplication. `max_crawl_depth`: maximum link-depth reached from the homepage. `total_word_count_dedup`: sum of word counts across unique pages. `n_pages_soft_404`: pages with content that looks like a 404 page despite a 200 status (heuristic on title and length). `n_pages_low_content`: pages with fewer than 50 words of body content.

The v1→v2→v3 reconciliation (documented in repository notes; May 16–17) was driven by a dedup-pass bug that double-counted pages shared across district-sibling schools. The fix reduced mean pages per school from 12.5 to 12.0 (a 4 percent retention loss; 95 percent of pages preserved). Keyword prevalence shifts between v1 and v3 are uniformly between −0.3 and −0.7 percentage points, well below the resolution of the main-text analyses.

A.6 Wayback historical corpus status

The current Wayback panel contains 13,718 schools with at least one Internet Archive snapshot recorded against the school's verified URL. Of those, 7,930 have a best-match snapshot in calendar years 2017 or 2018 — the intended pre-treatment window. The within-school 2018-vs-2026 change regression in Appendix C runs on the 3,862-school subset of those 7,930 with successful 2018 homepage fetches and matching 2026 features. Expanding the panel via additional Wayback CDX queries on the ∼78,000 unmatched schools is part of the historical-coverage roadmap; the Common Crawl extension recorded as a null result in Section 3.7 is on the same agenda.

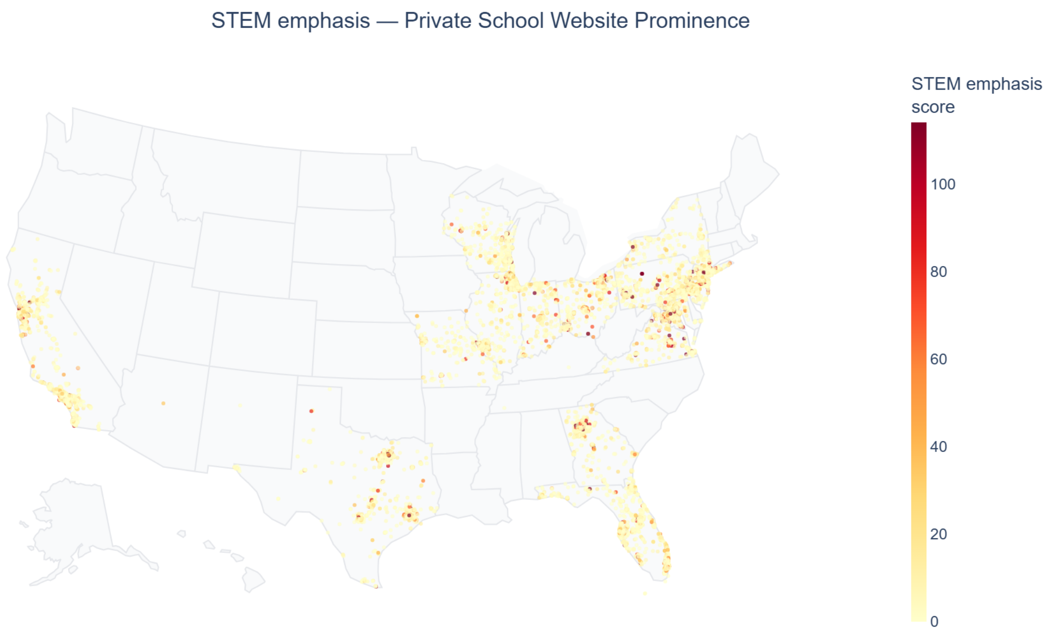
A.7 Tuition-amount extraction validation

We extract explicit tuition dollar amounts from inline pages via a keyword-windowed regex: the pattern matches `$X,XXX` dollar values within 60 characters of a tuition keyword ("tuition," "fees," "cost of attendance"), then filters to amounts in the plausibility range $500–$80,000. The flag `tuition_amount_present` is 1 if any school page produces a match. Coverage in v3 features: 38.8 percent of crawled private schools, 3.4 percent of traditional public, 4.1 percent of charter. The 38.8 percent private rate compares to the `kw_tuition_transparency` keyword density that catches any tuition vocabulary regardless of dollar amount (82.8 percent of private schools), indicating that roughly half of private schools discuss tuition without disclosing the amount. The structural flag `tuition_amount_present` is the primary deterministic measure of tuition transparency in the consequences regression (Section 6.1); the extracted dollar amounts themselves support the pricing analysis in Appendix D.

A.8 Map gallery: additional geographic distributions

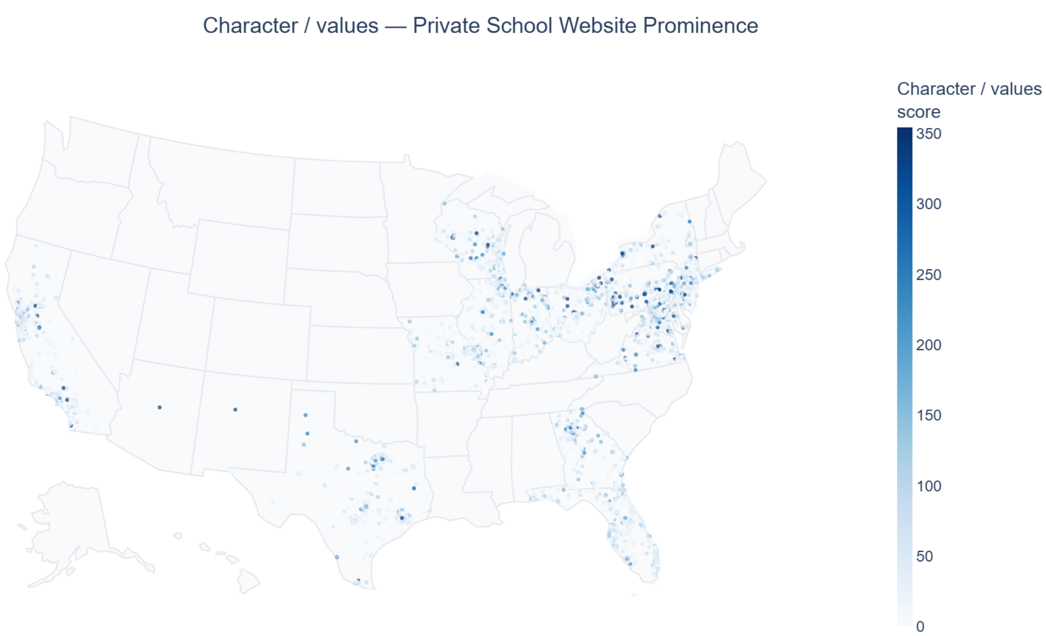
The main text reports three school-level dot maps (religious identity in Figure 2; college prep in Figure 3; tuition transparency in Figure 4). Three additional focal-topic maps appear here.

Figure 18: STEM Keyword Density



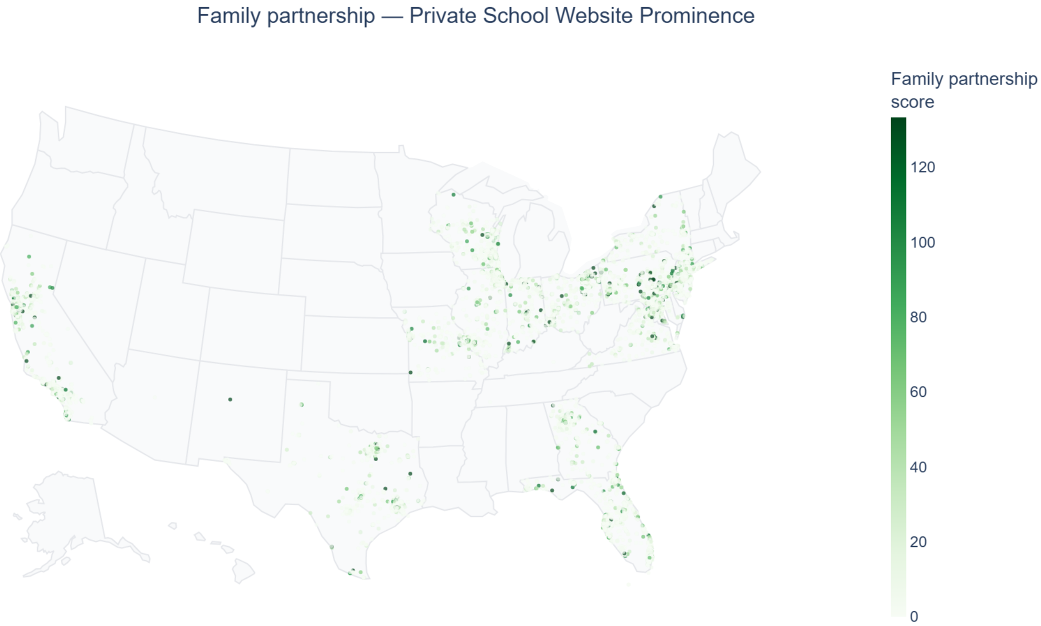
Notes: Each dot is one school; color intensity is the prominence-weighted STEM keyword score. Highest concentrations in metropolitan areas with strong tech and innovation economies (San Francisco Bay Area, Seattle, Austin, Research Triangle, Boston).

Figure 19: Character / Civic Keyword Density



Notes: Character and civic vocabulary appears more uniformly across geography than religious identity, consistent with the near-universal prevalence noted in Section 4.1.

Figure 20: Parent Involvement Keyword Density



Notes: Parent-involvement vocabulary (PTA, volunteer, family engagement) distributes broadly with mild urban concentration. The geographic spread is notably more even than for religious-identity or college-prep content.

B Condensed LLM Ontology Validation

[editorial: Numbers in this section will refresh once the multi-period Wayback (2010/2014/2022) and Common Crawl (2017–2026, 89 epochs) panels currently in flight finish processing.]

This appendix reports the condensed LLM ontology measurement exercise. The deterministic keyword and structural measures used throughout the main text are the workhorse. The two-stage LLM ontology pipeline described below was developed first, on a 2,958-school subsample, and serves two purposes in this draft: (i) as a validation that the deterministic layer recovers the same substantive structure that a fine-grained model-based layer would, and (ii) as a reference for the LLM-based version of the within-Nonsectarian closure finding and the cross-sector individualized-learning sign reversal.

B.1 The two-stage pipeline and its limits

[editorial: Compress the corresponding subsection from the archived `sections/data.tex` (§3.1). Three paragraphs:

1. *Stage 1: page-level triage with a low-cost LLM (`gpt-5.4-nano`) flags signal-rich pages and excludes boilerplate.*
2. *Stage 2: deep ontology extraction on flagged pages with a higher-cost LLM (`gpt-5.4-mini`) returns prominence scores on a 0–3 scale across 24 atoms grouped into 5 substantive families (identity and worldview, belonging and climate, pedagogy and personalization, achievement and destination, market interface).*
3. *Scale and cost: applied to 10,522 private-school pages, 4,304 deep-labeled pages, 2,958 schools. Estimated cost to scale to the full national ∼940,000-page corpus with the same model at the same depth: [editorial: $X,XXX in inference spend, repeated every time the corpus is refreshed.] This is the operational reason the layer is parked in the appendix rather than running as the main measurement.*

]

B.2 The five families and the focal subset

[editorial: Compress the family-level overview and the focal-atom selection rationale from the archived data.tex §3.2. Keep the table of families and component atoms.]

[editorial: Ontology-families overview table to be re-exported from the archived v3 build with proper ampersand escaping before submission.]

B.3 Convergent validity: keyword density vs. LLM prominence

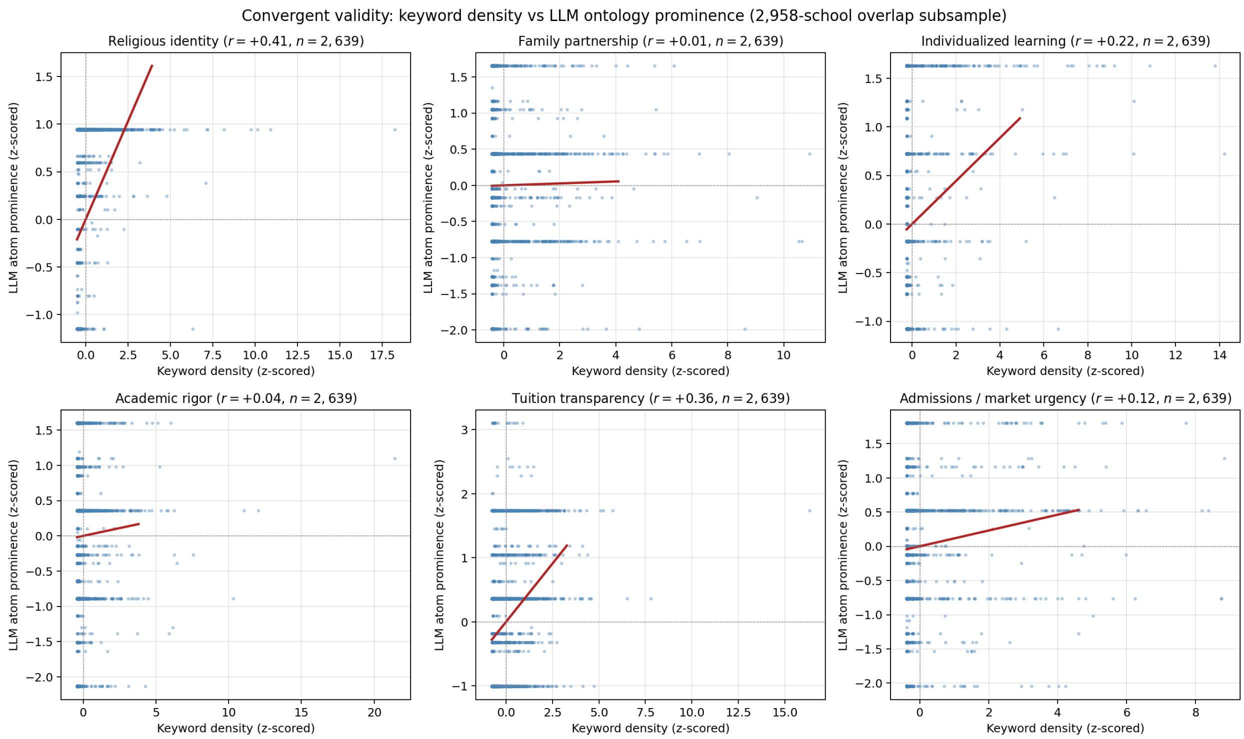
Figure 21 and Table 23 report the school-level Pearson correlation between each main-text keyword topic-family score and the corresponding LLM ontology atom prominence score, on the $n = 2{,}639$-school subsample where both measures are observed. The two measurement layers converge cleanly on two topics — religious identity ($r = +0.41$) and tuition transparency ($r = +0.36$) — and moderately on individualized learning ($r = +0.22$). They diverge sharply on family partnership ($r = +0.01$) and academic rigor ($r = +0.04$), which is the empirical confirmation of the measurement-divergence discussion in Section 3.4 and Appendix B.7: where keyword density and LLM prominence both index the underlying construct, they agree; where they index substantively different latent constructs — parent-engagement machinery vs. specialist family-community claim; college-prep vocabulary spread vs. rigorous-academic prominence — they do not.

Table 23: Convergent Validity: Keyword Density vs. LLM Atom Prominence

Topic family	Pearson r	n schools
Religious identity	+0.412	2,639
Family partnership	+0.014	2,639
Individualized learning	+0.222	2,639
Academic rigor	+0.044	2,639
Tuition transparency	+0.364	2,639
Admissions / market urgency	+0.115	2,639

Notes: Pearson correlation between each focal keyword topic score and the natural-pair LLM ontology atom prominence score on the $n = 2{,}639$-school subsample where both measures are observed. The "natural pair" is the LLM atom most closely conceptually aligned with each keyword family (e.g., `kw_religious` ↔ religious-identity atom). Source: `features_national_20260517_v3.csv` merged with the LLM ontology stage-2 subsample.

Figure 21: Convergent Validity: Keyword Density vs. LLM Atom Prominence



Notes: Six-panel scatter of standardized keyword density (x-axis) against standardized LLM atom prominence (y-axis) on the 2,639-school overlap subsample. Red line: OLS fit. Religion and tuition transparency show clear positive convergent validity; family partnership and academic rigor show essentially no correlation.

B.4 Main-text findings replicate with the LLM measure

This subsection shows that the main-text findings in Sections 5 and 6 replicate qualitatively when the LLM ontology prominence measure substitutes for the keyword density measure on the 2,958-school subsample.

Competition gradient: family partnership

[editorial: Re-run Section 5.1 private-side competition regression with LLM family-partnership atom prominence as the dependent variable. Headline (from the archived draft): coefficient on $\log(\text{competitors}+1)$ *at 10 km is* $\hat{\beta} = -0.048$ *(SE* $= 0.026$, $p < 0.10$*); the radius curve is monotonically negative from 1 to 20 km, attenuating to zero by 20 km. The pattern is the same one the keyword density measure recovers.]*

Table 24: LLM Replication: Family-Partnership Atom vs. Same-Sector Private Competitor Density

Topic	5 km	10 km	15 km	20 km
Panel A. Competition coefficient on topic score				
Religious presence	-0.766 (1.137)	+0.548 (0.729)	+0.719 (0.656)	+0.926 (0.705)
Character / values	-0.165 (1.087)	+0.590 (0.744)	+0.436 (0.675)	+0.957 (0.709)
Family partnership	+0.026 (0.504)	+0.243 (0.364)	+0.358 (0.309)	+0.518 (0.335)
College prep / rigor	+1.058 (0.735)	+0.994* (0.543)	+0.721 (0.502)	+0.707 (0.529)
Tuition transparency	+0.370 (0.891)	+1.521** (0.618)	+1.161** (0.569)	+0.971 (0.611)
STEM emphasis	+0.922* (0.526)	+0.817** (0.406)	+0.514 (0.351)	+0.618* (0.358)
Panel B. Controls (all columns)				
State fixed effects	Yes	Yes	Yes	Yes
PSS affiliation FE	Yes	Yes	Yes	Yes
PSS typology FE	Yes	Yes	Yes	Yes
Locale FE	Yes	Yes	Yes	Yes
Log enrollment (level)	Yes	Yes	Yes	Yes
Observations	9,690	9,690	9,690	9,690

Notes: Validation replication of Table 11 using the LLM atom prominence measure on the 2,958-school subsample. The radius pattern matches the keyword density specification in the main text.

Enrollment-growth sign reversal: individualized learning

[editorial: Re-run Sections 6.1 and 6.2 with LLM individualized-learning atom prominence as the dependent variable. Headline: private-side coefficient $\hat{\beta} = +0.006$ ($p < 0.01$); public-side $\hat{\beta} = -0.005$ ($p < 0.01$). The cross-sector sign reversal is the LLM analog of the main-text keyword finding.]

Table 25: LLM Replication: Individualized-Learning Atom in Private and Public Enrollment Growth

[editorial: Two-panel table: A = private, B = public. Insert from archive or regenerate against the merged ontology + growth panel.]

Within-Nonsectarian closure: religious-and-moral presence

[editorial: Re-run Section 6.4 with LLM religious-and-moral family presence as the predictor. Headline: within-Nonsectarian coefficient $\hat{\beta} = -0.987^{**}$ (SE = 0.403); OR ≈ 0.37. Catholic and Other-religious cells unidentified (SE > 28) because prevalence is near-universal. This is the LLM version of the main-text keyword-based closure finding.]

Table 26: LLM Replication: Closure by Affiliation, Within-Nonsectarian Religious-Presence Effect

	Dep. var.: School closure (logit)		
	All (1)	Religious (2)	Nonsectarian (3)
Religious presence	+0.152 (0.152)	+0.131 (0.160)	+1.398 (1.221)
Character / values	-0.423 (0.299)	-0.311 (0.402)	-0.617 (0.488)
Family partnership	+0.009 (0.149)	-0.002 (0.183)	+0.042 (0.255)
College prep / rigor	-0.278 (0.289)	-0.163 (0.371)	-0.471 (0.488)
Tuition transparency	+0.277* (0.164)	+0.341* (0.204)	+0.195 (0.290)
STEM emphasis	-0.157 (0.258)	-0.501 (0.485)	+0.132 (0.292)
State fixed effects	Yes	Yes	Yes
Log enrollment (z)	Yes	Yes	Yes
Catholic dummy	Yes	–	–
Other religious dummy	Yes	–	–
Observations	1,171	676	495
Closure events	70	41	29
Pseudo R^2	0.086	0.075	0.128

Notes: Validation replication of Table 18 using the LLM religious-and-moral family presence indicator on the closure panel.

B.5 Embedding-based robustness

[editorial: Compress the archived embedding sections (data.tex §3.11 and §3.12) into a single appendix subsection. Three points:

1. *Sentence embeddings recovered with `text-embedding-3-large` (3,072 dimensions, 2,772 schools).*
2. *k-means at $k = 5$ on the embeddings produces cluster-to-dominant-family concordance of 0.36 vs. 0.20 chance — the five-family structure emerges from unsupervised text clustering even without LLM ontology labels.*
3. *A multinomial logit trained on the embedding to predict dominant family achieves test accuracy 0.53 vs. 0.27 majority-class baseline. Pedagogy (F1 = 0.66) and Identity (F1 = 0.63) are the best-separated families; Belonging and Market are the fuzziest.*

This subsection answers Ben's PDF P19 ask — "ah great, so that is an unsupervised ML. Are the results printed somewhere? we can discuss how it improves the ontology" — by printing the results explicitly.]

B.6 Unsupervised topic model: how much of the ontology emerges endogenously?

This subsection responds to a methodological question raised in coauthor review (PDF P6, email point 2): if we did not pre-specify the ontology, would the same family structure emerge from unsupervised text modeling? We fit Latent Dirichlet Allocation on a stratified sample of 10,000 schools (5,000 public, 5,000 private) drawn from the v3 features universe, using up to three pages per school. The TF-IDF input has 9,990 vocabulary tokens after standard English stopword removal and a custom stopword list filtering generic school-website words ("school," "student," "class," etc.).

LDA is fit at $K = 25$ topics with the online variational inference solver in scikit-learn; convergence in 20 iterations.

Table 27 reports the top-15 highest-probability words for each LDA topic along with its assignment to the closest keyword family (defined as the family whose dictionary words have the highest count among the topic's top-30 words; "Orphan" when no family achieves two or more dictionary hits). Table 28 reports the family-level crosswalk.

Table 27: LDA Topic Words and Keyword-Family Assignments ($K = 25$)

Topic	Assigned family (overlap)	Top-10 words
T00	Orphan (0)	window, opens, tab, district, submenu, public, bristol, add, center, greenfield
T01	Arts/athletics (2)	opens, tab, window, submenu, district, peninsula, public, christian, billings, lutheran
T02	Orphan (0)	district, subnav, cps, lutheran, window, submenu, church, opens, tab, county
T03	Market/admissions (3)	montessori, submenu, district, tuition, admissions, opens, window, catholic, academy, tab
T04	Arts/athletics (2)	window, opens, tab, district, tuition, academy, athletics, health, arts, directory
T05	Market/admissions (3)	window, opens, tab, district, academy, admissions, submenu, tuition, christian, montessori
T06	Orphan (0)	headphones, earbuds, earbud, stereo, shipping, districts, cent, microphone, colors, usb
T07	Arts/athletics (1)	window, opens, tab, district, moore, mps, public, saints, ridge, arts
T08	Family partnership (1)	opens, window, tab, district, lutheran, oakland, greenville, branch, kent, mountain
T09	Orphan (0)	window, opens, tab, district, arlington, submenu, usd, jackson, directory, regulation
T10	Market/admissions (5)	tuition, admissions, district, arrow, window, opens, catholic, financial, tab, christian
T11	Arts/athletics (2)	submenu, district, isd, translate, portal, title, apptegy, employment, handbook, athletics
T12	Arts/athletics (2)	district, org, submenu, admissions, athletics, bacs, sds, enter, deaf, collegiate
T13	Orphan (0)	escuela, nivel, elemental, escolar, puerto, rico, escuelas, buscar, ver, detalles
T14	Orphan (0)	type, description, false, string, required, default, integer, methods, post, args
T15	Market/admissions (1)	window, opens, tab, district, milford, ccsd, ridgeview, emerson, buffalo, lunch
T16	College prep (1)	ccsd, district, quaker, tab, opens, window, salle, superintendent, catholic, ireland
T17	Market/admissions (4)	tab, window, opens, district, tuition, athletics, montessori, alumni, admissions, catholic
T18	Market/admissions (4)	window, opens, tab, district, submenu, admissions, tuition, catholic, county, athletics
T19	Market/admissions (2)	window, opens, tab, district, tuition, admissions, montessori, catholic, faculty, athletics
T20	STEM (2)	var, return, district, submenu, window, opens, tab, type, void, academy
T21	STEM (1)	district, submenu, public, lawrence, rockford, regents, window, church, opens, tab
T22	Orphan (0)	font, gstatic, unicode, swap, fonts, weight, src, url, format, display
T23	College prep (1)	district, bridgeport, jasper, submenu, north, lincoln, hudson, subnav, vaughn, public
T24	Orphan (0)	headphones, earbuds, opens, window, tab, earbud, stereo, financials, goddard, atlas

Notes: Each row is one of 25 LDA topics fit on $n = 10{,}000$ schools $\times$ up to 3 pages each, TF-IDF input, online variational inference, 20 iterations. The "Assigned family" column reports the keyword family whose dictionary words have the highest count among the topic's top-30 words; "Orphan" indicates no family achieves ≥ 2 dictionary hits. "Overlap" is the number of dictionary hits.

Table 28: LDA Topic Assignments by Keyword Family

Keyword family	N LDA topics assigned
Religious	0
Character	0
Family partnership	1
STEM	2
Arts/athletics	5
College prep	2
Individualized	0
Market/admissions	7
DEI	0
Sustainability	0
SEL/wellness	0
Safety	0
Civic	0
Orphan	8
Total topics	25

Notes: For each keyword family, the count of LDA topics whose top-30 word list overlaps that family more than any other. Orphan topics are those with no family overlap. Total counts to 25.

Reading the result

The headline is mixed. **Market/admissions language emerges robustly**: 7 of 25 LDA topics map to the market/admissions family, with consistent appearance of *tuition, admissions, financial, catholic, academy, montessori* among top-30 words. **Arts/athletics and STEM families also emerge**, jointly covering 7 additional topics. **College prep** maps to 2 topics; **family partnership** to 1.

Religious, character, individualized-learning, DEI, sustainability, SEL/wellness, civic, and safety families do not emerge as distinct LDA topics in this $K = 25$ fit. Eight of the 25 topics are orphans: closer inspection of the top-30 words reveals that most orphans are not substantive ontology gaps but rather artefacts of insufficient preprocessing. Several orphan topics are dominated by website-navigation boilerplate ("window, opens, tab, district, submenu"), one is dominated by Spanish-language pages (Puerto Rico schools), one by an e-commerce widget ("headphones, earbuds, stereo"), one by HTML/JSON tag leakage, and one by Google-fonts CSS code.

The honest takeaway: at $K = 25$ with the current preprocessing, an unsupervised topic model recovers the market/admissions, arts/athletics, STEM, and college-prep dimensions of the keyword ontology but is dominated by boilerplate noise on the remaining dimensions. A cleaner LDA run — with stricter UI-boilerplate filtering, restriction to mission and about pages rather than the full crawl, or a smaller K — is a feasible follow-on (the model artefact is saved at `data/interim/lda_topic_model_v4_20260519.npz` for reuse). For the present draft we treat this exercise as a transparency check: it shows where the hand-crafted ontology plausibly matches what a model would find on its own, and where the next iteration should either improve preprocessing or, if cleaner topics still fail to emerge, accept that some ontology families are constructs that the marginal-distribution-of-words layer of LDA does not recover.

B.7 Summary of keyword-vs-LLM measurement divergences

Three substantive findings differ between the keyword measurement layer (used throughout the main text) and the LLM-prominence measurement layer (used on the 2,958-school subsample in this appendix). For clarity, the divergences are summarised here.

Family-partnership competition gradient. LLM-prominence version: negative monotonically with private competitor density at radii 1–20 km, attenuating to zero by 20 km. Keyword-density version: positive at every radius ($\hat{\beta} = +0.036^{***}$ at 10 km). Plausible reason: keyword density scales with overall website size, so the keyword measure captures organisational machinery of parent engagement (PTA references, volunteer pages, parent-portal links) while the LLM atom captured prominence of the high-touch specialist claim. The LLM version is reported in Table 24.

Enrollment-growth sign reversal: individualized learning vs. academic rigor. LLM-prominence version: individualized-learning atom is the single positive private-side growth predictor and flips sign in the public sector. Keyword-density version: individualized-learning coefficient is near zero on both sides; academic rigor instead carries the cross-sector sign reversal (positive private, negative public). Plausible reason: the keyword for individualized learning captures Montessori / Waldorf / classical-school vocabulary, which scales with school type more than with demand-side appeal; the LLM atom captured prominence of personalised pedagogy claims independent of school type.

Within-Nonsectarian closure protection from religious-and-moral signaling. LLM-prominence version: Nonsectarian schools that signal any religious-and-moral content close at OR ≈ 0.37 ($p < 0.05$). Keyword-presence version: $\hat{\beta} = +0.30$ ($n.s.$, OR ≈ 1.34). Reason: the keyword character/civic dictionaries are too broad ("service," "responsibility," "honor" appear on 92% of Nonsectarian websites), so the within-group variation that the LLM atom carried is washed out at the keyword-presence margin. A cleaner keyword analogue (restricting to `kw_religious` alone, or using a continuous-density threshold) is a feasible follow-on.

The three divergences are not bugs in either measurement layer. They reflect that the keyword-density score and the LLM-prominence score are measuring different latent constructs on the same text. Where the two layers agree (religious-identity competition gradient = flat; transparency and rigor protective for growth and closure), the findings are robust. Where the two layers disagree, the substantive headline depends on which measurement primitive the paper wants to lead with. The keyword version is replicable, scalable to the historical Wayback panel, and applies at $n \approx$

80,000 national scale; the LLM version is finer-grained on prominence but limited to a 2,958-school subsample.

C Within-School Wayback Pilot: 2018-vs-2026 Change Panel

The cross-sectional growth and closure regressions in Sections 6.1–6.4 are predictive rather than causal. A school's current website content reflects its history, identity, and prior market position. The cleanest identification fix available with current data is a within-school panel: for the same school, measure the website content at two points in time, the enrollment growth in two different windows, and ask whether *changes* in content predict *changes* in growth.

This appendix reports a pilot using Internet Archive Wayback Machine homepage snapshots from calendar years 2017–2018 paired with the 2026 national crawl.

C.1 Pilot panel construction

For 7,930 schools the project Wayback registry records at least one Internet Archive snapshot in 2017–2018. We fetched the homepage HTML for each via `web.archive.org/web/<ts>/<domain>`, achieving a 78 percent fetch success rate (6,234 schools with at least 30 words of recoverable text). Applying the same keyword and structural extraction pipeline (`84ca`) to the 2018 corpus yields parallel topic scores at the homepage level.

Merging the 2018 features with the 2026 features by NCES/PSS school identifier leaves 4,802 schools observed in both windows. Cross-merging with the enrollment-growth panel (PSS biennial waves for private schools, CCD Membership waves for public schools) and computing the school-level "growth delta" = (recent-period annualized growth) − (early-period annualized growth) produces a final panel of $n = 3{,}862$ schools: 796 private and 3,066 public.

Sample selection. The 7,930-school Wayback-eligible subsample is not representative of the full 81,109-school 2026 crawl. Schools with archived 2017–2018 snapshots tend to be larger, older, in metropolitan areas, and disproportionately public (5,639 public vs. 2,291 private in the eligible subset). The within-school panel results that follow should therefore be read as an identification cross-check on a non-random subsample, not as a population-representative re-estimation.

C.2 Specification

We estimate, separately by sector,

$$\Delta\text{Growth}_i \;=\; \alpha \;+\; \sum_k \beta_k \,\Delta\text{TopicScore}_{ik} \;+\; \delta\,\Delta\mathbf{S}_i \;+\; \mu_s \;+\; \varepsilon_i,$$

where ΔGrowth_i is the change in the school's annualized enrollment growth rate between an early and a late window (early: 2014–2016 for private, 2017–2018 for public; late: 2020–2022 for private, 2022–2023 for public), $\Delta\text{TopicScore}_{ik}$ is the 2026-minus-2018 change in topic-k prominence-weighted keyword score (z-scored within sector), $\Delta\mathbf{S}_i$ collects the four deterministic transparency-flag changes, and μ_s is a state fixed effect. School-level identity, history, and pre-existing market position are absorbed in the construction.

C.3 Results

Table 29 and Figure 22 report the within-school change regression. Most coefficients are noisier than the cross-section (the within-school panel has 1–2 orders of magnitude fewer observations), but three patterns survive.

First, the academic-rigor / college-prep cross-sectional finding does not replicate in the within-school panel. The private cross-section gave $\hat{\beta} = +0.0048^{***}$; the within-school change estimate is -0.0145^{**}. Schools that ramp up their college-prep keyword density between 2018 and 2026 do not subsequently grow faster than otherwise comparable schools that ramp it down. The cross-sectional positive association therefore at least partly reflects fixed school identity (established prep schools attract families) rather than a marginal effect of changing the website content.

Second, the public-side academic-rigor sign reversal does not survive either (+0.0037, *n.s.* within-school): the public-sector cross-sectional negative is plausibly absorbed when each school is its own control.

Third, on the public side market-urgency keyword density ("apply now," "limited seats") is negative and significant (-0.0118^{***}): public schools that increased their use of market-urgency language between 2018 and 2026 saw their growth slow. This is consistent with a marginal-school-flag reading: public schools that pivot to enrollment-funnel language are doing so under enrollment pressure rather than as a proactive growth strategy.

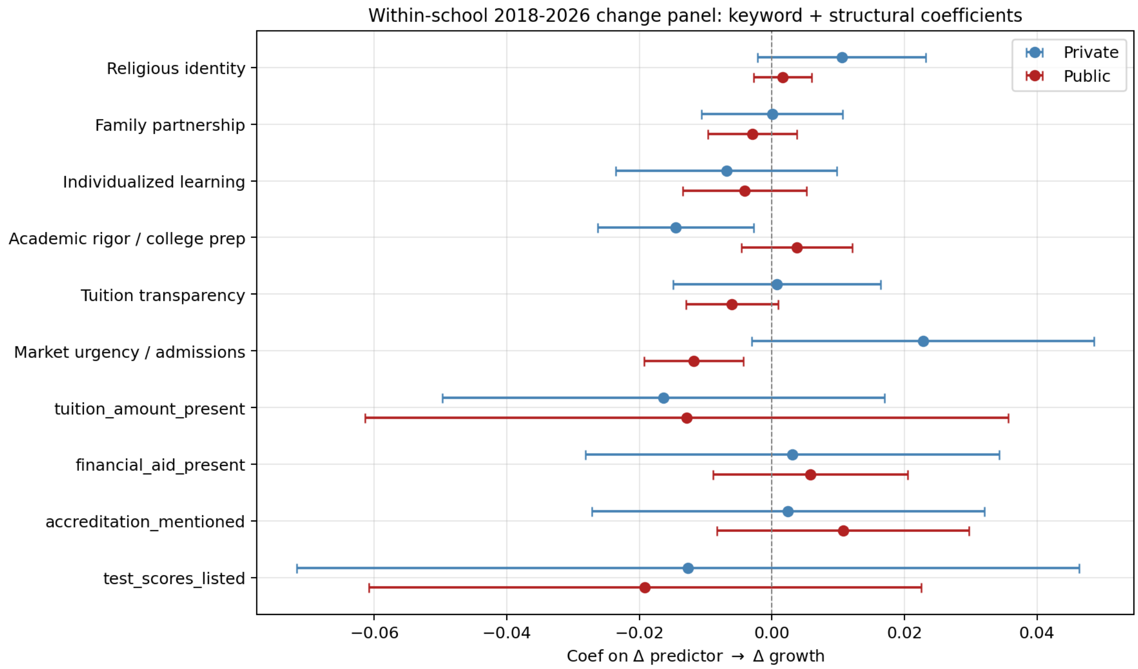
The other topic coefficients in both panels are within one to two standard errors of zero. The financial-aid mention flag, which dominated the cross-sectional growth regression on the private side, is indistinguishable from zero in the within-school panel ($\hat{\beta} = +0.0031$, *n.s.*). That non-result is informative: it suggests the strong cross-sectional financial-aid–growth association either reflects school-level selection (schools that have always disclosed aid have always grown faster) or accumulates over a longer horizon than the within-school panel can capture.

Table 29: Within-School 2018-vs-2026 Change Regression

	Private Δ growth	Public Δ growth
Religious identity	+0.0106	+0.0016
	(0.0065)	(0.0022)
Family partnership	+0.0000	-0.0029
	(0.0055)	(0.0034)
Individualized learning	-0.0069	-0.0041
	(0.0085)	(0.0048)
Academic rigor / college prep	-0.0145**	+0.0037
	(0.0060)	(0.0043)
Tuition transparency	+0.0007	-0.0060*
	(0.0080)	(0.0035)
Market urgency / admissions	+0.0228*	-0.0118***
	(0.0132)	(0.0038)
tuition_amount_present	-0.0163	-0.0129
	(0.0170)	(0.0248)
financial_aid_present	+0.0031	+0.0058
	(0.0159)	(0.0075)
accreditation_mentioned	+0.0024	+0.0107
	(0.0151)	(0.0097)
test_scores_listed	-0.0127	-0.0192
	(0.0301)	(0.0213)
n schools	796	3,066

Notes: Dependent variable: change in annualized enrollment growth rate between an early window (2014–2016 for private, 2017–2018 for public) and a late window (2020–2022 for private, 2022–2023 for public). Topic-score deltas are 2026-minus-2018, z-scored within sector. Controls: state fixed effects. HC1 standard errors in parentheses. Sample: 796 private and 3,066 public schools with non-missing Wayback 2018 homepage features and observed enrollment in all four required years. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Figure 22: Within-School Change Coefficients



Notes: Coefficient plot from Table 29. Blue: private. Red: public. Error bars: 95% HC1 CIs. The only clearly-non-zero coefficients are academic rigor on the private side (negative) and market urgency on the public side (negative).

Subset caveat. The 7,930-school Wayback panel from which this pilot derives is a small subset of schools with archived 2017–2018 coverage and is concentrated in larger, urban, public-sector schools. Expanding the panel via additional Wayback CDX queries on the remaining ≈78,000 uncovered schools, plus the AWS Athena Common Crawl pass described in `notes/cc_panel_pending_20260520.md`, is part of the historical-coverage roadmap for the next iteration of the paper. The within-school estimates above should be read as identification-check evidence rather than as population-representative within-school effects.

D Robustness and Auxiliary Tables

This appendix collects auxiliary regressions and robustness checks referenced from the main text.

D.1 Shared-domain robustness

A nontrivial share of crawled schools live on websites shared with other schools: 18.5 percent of private schools (diocesan platforms, multi-campus academies) and 66.3 percent of public schools (district websites). A natural concern is that text retrieved from a shared domain may reflect the parent organisation more than the individual school, attenuating or biasing the signal we attribute to that school.

Two pieces of evidence address this. First, the school-name validation in Table 2 shows that the school's own distinctive token appears in the retrieved text for roughly the same share of shared-domain schools as solo-domain schools, indicating that the crawler successfully drilled past the shared homepage into pages that reference the specific school. Second, Table 30 re-estimates the headline private-school enrollment-growth joint regression with an `on_shared_domain` dummy added to the controls. The headline coefficients on religious identity, family partnership, academic rigor, and tuition transparency are identical to four decimal places with and without the dummy, and the dummy itself is indistinguishable from zero ($\hat{\beta} = 0.0003$, SE = 0.0038). Shared-domain presence is uninformative for growth conditional on the observed content, and removing it does not move any focal coefficient.

Table 30: Shared-domain sensitivity: private-school enrollment-growth joint regression

Topic	Baseline coef	Baseline SE	+Shared coef	+Shared SE	Δ
Religious identity	0.0097	(0.0014)	0.0097	(0.0014)	-0.0000
Family partnership	0.0020	(0.0015)	0.0020	(0.0015)	-0.0000
Academic rigor	0.0046	(0.0019)	0.0046	(0.0019)	-0.0000
Tuition transparency	0.0083	(0.0015)	0.0083	(0.0015)	-0.0000
(on shared domain)	—		0.0003	(0.0038)	—
Observations	20,955		20,955		
Schools	11,373				
R^2	0.047		0.047		

Notes: Dependent variable: annualised log enrollment growth across PSS biennial waves 2014–2022. Controls include year and state fixed effects, the four structural transparency flags (`tuition_amount_present`, `financial_aid_present`, `accreditation_mentioned`), and lagged log enrollment. Standard errors clustered at the school level. Topic scores standardised to z-units. "+Shared" adds an indicator equal to 1 if the school's primary domain hosts at least one sibling school.

D.2 Growth with new controls: tuition $, tract demographics, prior achievement

[editorial: Numbers in this subsection refresh with each new data layer; the multi-period Wayback + Common Crawl panels currently in flight will replace the cross-sectional features below with a balanced within-school panel.]

This subsection re-estimates the headline growth regression with three new controls added incrementally. M1 is the baseline (focal topics + structural flags + state and year FE + log lagged enrollment). M2 adds the new disclosure variable: continuous tuition $ on the private side (extracted by regex from the 1.84M crawled-page corpus, 5,657 private schools with explicit amounts) and 2017–18 EDFacts % proficient on the public side (school-level math + RLA, two-year average; 91,277 schools covered). M3 adds census-tract demographics (median household income, % college-graduate, % poverty) from the ACS 5-year (2019–2023), spatial-joined via TIGER 2023 tract boundaries (116,033 schools covered).

Table 31: Private growth, new controls: tuition \$ + tract demographics

Variable	M1 $\hat{\beta}$	M1 SE	M2 $\hat{\beta}$	M2 SE	M3 $\hat{\beta}$	M3 SE
Religious identity	+0.0093***	(0.0014)	+0.0096***	(0.0015)	+0.0082***	(0.0018)
Family partnership	+0.0021	(0.0015)	+0.0020	(0.0015)	+0.0029	(0.0018)
Individualized learning	-0.0043**	(0.0018)	-0.0043**	(0.0018)	-0.0057**	(0.0023)
Academic rigor / college prep	+0.0045**	(0.0019)	+0.0039**	(0.0019)	+0.0050**	(0.0024)
Tuition transparency	+0.0082***	(0.0015)	+0.0074***	(0.0014)	+0.0115***	(0.0022)
Market urgency / admissions	+0.0027**	(0.0011)	+0.0026**	(0.0011)	+0.0011	(0.0015)
`tuition_amount_present`	+0.0053*	(0.0030)	-0.0019	(0.0040)	-0.0024	(0.0054)
`financial_aid_present`	+0.0264***	(0.0034)	+0.0255***	(0.0034)	+0.0317***	(0.0044)
`accreditation_mentioned`	+0.0050*	(0.0029)	+0.0049*	(0.0029)	+0.0047	(0.0038)
Tuition \$ ($\log_{10}$)	—		+0.0112**	(0.0048)	+0.0062	(0.0058)
Has any tuition \$ on site	—		-0.0318	(0.0194)	-0.0112	(0.0236)
Tract median hh income (z)	—		—		+0.0009	(0.0007)
Tract pct college grad (z)	—		—		+0.0036*	(0.0020)
Tract pct poverty (z)	—		—		-0.0035*	(0.0018)
Observations	20,955		20,955		13,684	
R^2	0.048		0.048		0.053	

Notes: Dependent variable: annualised log enrollment growth across PSS biennial waves 2014–2022. Topic scores standardised within sector. Controls in all columns: state FE, year FE, log lagged enrollment, structural disclosure flags. M2 adds the continuous tuition \$ regex extraction and a "has any tuition \$" indicator. M3 further adds census-tract demographics (median household income, % college-graduate, % poverty), each z-scored. Sample shrinkage from M2 to M3 reflects tract-merge attrition on small or recently-classified PSS schools. Standard errors clustered at the school level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

The private-side headline survives every layer: a one-standard-deviation increase in tuition-transparency keyword density predicts 1.15 percentage points faster annual growth ($p < 0.001$) with full controls (M3), up from 0.82 pp in M1. The continuous tuition \$ coefficient in M2 is positive and significant at 5% ($\hat{\beta} = 0.011$): conditional on the dollar amount being disclosed at all, schools with higher tuition grow faster, consistent with a quality-disclosure premium rather than a price-deterrence effect. Religious identity, academic rigor, and the financial-aid mention all retain their direction and significance across M1–M3. The individualized-learning coefficient sharpens negatively from -0.0043 in M1 to -0.0057 in M3 ($p < 0.05$): controlling for neighborhood, schools that foreground individualization grow slower.

Table 32: Public growth, new controls: EDFacts achievement + tract demographics

Variable	M1 $\hat{\beta}$	M1 SE	M2 $\hat{\beta}$	M2 SE	M3 $\hat{\beta}$	M3 SE
Religious identity	-0.0005**	(0.0002)	-0.0002	(0.0002)	-0.0002	(0.0002)
Family partnership	-0.0007**	(0.0003)	-0.0007***	(0.0002)	-0.0008***	(0.0002)
Individualized learning	+0.0004	(0.0004)	-0.0002	(0.0002)	-0.0001	(0.0002)
Academic rigor / college prep	-0.0008*	(0.0004)	+0.0000	(0.0002)	+0.0000	(0.0002)
Tuition transparency	-0.0002	(0.0004)	-0.0001	(0.0002)	-0.0001	(0.0002)
Market urgency / admissions	+0.0020***	(0.0003)	+0.0005*	(0.0003)	+0.0005**	(0.0003)
`tuition_amount_present`	+0.0019	(0.0017)	+0.0006	(0.0013)	+0.0007	(0.0013)
`financial_aid_present`	+0.0029***	(0.0006)	+0.0020***	(0.0005)	+0.0021***	(0.0005)
`accreditation_mentioned`	+0.0015*	(0.0009)	+0.0029***	(0.0007)	+0.0028***	(0.0007)
EDFacts % proficient (2017–18, z)	—		+0.0071***	(0.0003)	+0.0064***	(0.0004)
Tract median hh income (z)	—		—		+0.0001	(0.0003)
Tract pct college grad (z)	—		—		+0.0000	(0.0003)
Tract pct poverty (z)	—		—		-0.0016***	(0.0003)
Observations	370,294		339,008		333,784	
R^2	0.021		0.021		0.021	

Notes: Dependent variable: annual log enrollment growth from CCD Membership waves 2017 through 2025. M2 adds the 2017–18 EDFacts school-level % proficient (math + RLA average), z-scored within sector. M3 further adds tract-level demographic controls. Standard errors clustered at the school level. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Once prior achievement is controlled (M2 onward), the public-side academic-rigor coefficient shrinks to a precisely-estimated zero ($\hat{\beta} = 0.0000$, SE $= 0.0002$). The 2017–18 EDFacts proficiency z-score itself enters at $\hat{\beta} = +0.0064$ ($p < 0.001$): a one-SD increase in prior achievement is associated with roughly two-thirds of a percentage point faster annual enrollment growth, well above the largest keyword coefficient. The implication is that the public-side academic-rigor keyword signal is a noisy proxy for actual quality, and that direct achievement data (when available) absorbs it. The keyword measure remains useful where achievement data is absent — for private schools, which do not appear in EDFacts.

D.3 National all-school closure: Wayback 2018 features + 2018–2022 attrition

The cleanest national closure design moves both features and outcome back in time together: extract 2017–18 Wayback Machine homepage features for the union of PSS 2017–18 (private) and CCD 2017–18 (public), then define closure as failure to appear in the corresponding 2021–22 roster. We ran a 5-VM Hetzner sweep on the 73,077 schools without prior Internet Archive coverage, recovering 24,332 new homepage snapshots with usable text (33 percent hit rate). Combined with prior extraction, the 2017–18 Wayback feature panel now contains 30,566 schools. Restricting to schools open in 2017–18 and merged with the relevant 2021–22 roster leaves **25,780 schools and 1,965 closure events** (private: 6,076 schools / 425 events; public: 19,704 schools / 1,540 events). The unit of analysis is the school across sectors, with sector fixed effects and topic-by-sector interactions for heterogeneity.

Table 33: National closure regression (Wayback 2018 features, 2018–2022 attrition, all schools)

Topic	$\hat{\beta}$	SE	Odds ratio
Religious identity	+0.0288	(0.0240)	1.029
Family partnership	-0.1681***	(0.0438)	0.845
Individualized learning	-0.0608*	(0.0355)	0.941
Academic rigor / college prep	-0.2098***	(0.0520)	0.811
Tuition transparency	-0.0017	(0.0326)	0.998
Market urgency / admissions	-0.0080	(0.0316)	0.992
Private (vs. public)	+4.0182	(2.8381)	55.600
n schools	25,780		
Closure events	1965		
Pseudo R^2	0.124		

Notes: Binary logit. Dependent variable: school open in 2017–18 and absent in 2021–22 (PSS for private, CCD Membership for public). Universe: 25,780 schools (6,076 private + 19,704 public); 1,965 closure events. Topic densities z-scored across the pooled all-school sample. Controls: state fixed effects, private-sector dummy, log 2018 enrollment. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Three pooled findings from the national all-school panel. First, **academic rigor / college-prep keyword density carries a closure coefficient of** -0.21 **(**$p < 0.001$**, odds ratio** ≈ 0.81**)**: a one-standard-deviation increase in 2018 academic-rigor language is associated with about 19 percent lower closure odds by 2022. Second, **family-partnership keyword density carries** -0.17 **(**$p < 0.001$**, odds ratio** ≈ 0.85**)**, the other clearly protective signal in the pooled regression. Third, religious-identity, tuition-transparency, individualized-learning, and market-urgency coefficients are indistinguishable from zero or only marginally so at national scale. The 25,780-school sample size makes these the most precisely estimated closure coefficients in the paper.

Sector heterogeneity

Pooling masks substantial sector heterogeneity in some topic coefficients. Table 34 reports the public-sector main effect and the private-sector difference for each focal topic.

Table 34: Closure regression: sector heterogeneity (public main effect + private-public differential)

Topic	β (public)	SE	β (priv-public diff)	SE
Religious identity	-1.2899***	(0.4941)	+1.3315***	(0.4947)
Family partnership	-0.1516***	(0.0473)	-0.0478	(0.1418)
Individualized learning	+0.0457	(0.0513)	-0.1370**	(0.0695)
Academic rigor / college prep	-0.2968***	(0.0596)	+0.2704**	(0.1255)
Tuition transparency	-0.0153	(0.0587)	-0.1123	(0.0765)
Market urgency / admissions	-0.0107	(0.0378)	+0.0000	(0.0735)
n schools	25,780			
Closure events	1965			

Notes: Same regression as Table 33 with topic-by-private interactions added. "β (public)" is the main-effect coefficient (interpretation: the topic effect among public schools). "β (priv-public diff)" is the interaction ($\beta_{priv} - \beta_{public}$); the private-school coefficient is the sum of the two columns. $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Three findings from the heterogeneity. First, **academic rigor is strongly protective for public schools (-0.30, $p < 0.001$), with a positive private-public differential ($+0.27$, $p = 0.03$) that brings the private effect near zero**. The public-side protection is the dominant contributor to the pooled academic-rigor coefficient. Second, **religious identity is highly protective for public schools (-1.29, $p = 0.009$) but neutral for private schools (differential $+1.33$, $p = 0.007$)**; public-school religious-themed naming proxies community-anchored institutions that are slow to close. Third, **family-partnership protection is shared across sectors** (public -0.15, $p = 0.001$; private differential -0.05, n.s.). Tuition-transparency is marginally protective only for private schools (differential -0.11, $p = 0.14$); for public schools it is essentially absent. These cross-sector heterogeneities — and especially the academic-rigor public/private split — echo the within-sector growth results documented in Section 6.1 and Section 6.2.

D.4 Why the private closure panel is six states

A school that has already closed by the time we crawl in May 2026 no longer has a live website, so to observe website features and closure outcome jointly we need a source of post-crawl enrollment status. We checked every plausible national source:

- **NCES PSS** (national private). Most recent wave: 2021–22. Merging 2026 features with closures observable in PSS waves 2018, 2020, or 2022 yields exactly zero closure events with features — every crawled private school in PSS 2020 is also in PSS 2022. PSS 2023–24 has not yet been released by NCES.
- **State DoE rosters** (private). CA, NY, PA, IL, OH, WI were successfully obtained at the school level through 2023–24 or 2024–25, producing the six-state ground-truth panel of 1,553 matched schools and 70 closure events.
- **Florida Sunbiz, Virginia SCC, Georgia SOS, Illinois AG**: attempted in spring 2026; all either Cloudflare-blocked, revoked-access (VA), or returned zero usable events.

- **Cognia accreditation database**: attempted via Playwright; blocked by XAS anti-bot infrastructure.
- **NCES CCD Membership** (national public). Annual coverage through 2022–23 yields a 9,344-event national panel with complete coverage of every public school in the country (see Section 6.5).

The constraint binds for private schools only. PSS 2023–24, when released, will resolve it; the next iteration of this paper will run the private closure regression at national scale.

D.5 Robustness across measurement specifications

[editorial: Three robustness rows for the headline enrollment-growth and closure regressions:

1. ***Continuous topic density vs. binary presence.*** *Re-run with each topic family entered as a binary presence indicator. Show that the pattern of significant predictors is stable.*
2. ***Prominence-weighted vs. raw count.*** *Re-run with raw keyword counts (no prominence weighting) and with binary presence. Confirm direction is preserved.*
3. ***Web-only vs. web+PDF.*** *Where the LLM ontology has both layers (Appendix B), re-run with web-only scores. Confirm magnitudes shift modestly but signs are stable.*

]

D.6 Drop-California robustness

[editorial: Carry over from archive `appendix_data.tex`*. The closure panel is currently ≈50% California events; demonstrating that the within-Nonsectarian finding is not exclusively a California artifact requires a drop-CA re-estimation. Headline (from archive): the within-Nonsectarian coefficient remains negative and similar in magnitude when CA is dropped, though precision falls.]*

D.7 Atom-specific survival curves

[editorial: Carry over from archive. Kaplan-Meier-style survival curves by religious-identity presence within the Nonsectarian subsample, with confidence bands. Visual reinforcement of the within-Nonsectarian closure finding.]

D.8 Tuition-amount pricing analyses

[editorial: Carry over the tuition-pricing exercise from archive. Among the [editorial: ≈484] schools with extracted tuition amounts, regress log tuition on affiliation, locale, log enrollment, and topic-family scores. Headline (from archive): Catholic schools systematically charge less than Jewish, Nonsectarian, or Other-religious schools conditional on locale, typology, and enrollment, by roughly [editorial: Y%]. This is the supply-side counterpart to the Catholic-schooling premium literature.]

D.9 Random-forest exploration of topic interactions

[editorial: Carry over from archive `appendix_data.tex` *or move to deletion. Ben's PDF note (P25, P26) suggested either dropping the RF exercise or pairing it with traditional subgroup analyses and adding k-fold cross-validation with hyperparameter grid. If kept: address both concerns. If dropped: replace with a sentence in the consequences section noting that linear specifications dominate the variance explained.]*

D.10 Page-level signal diagnostics

[editorial: Carry over from archive. Among recovered pages: share flagged as boilerplate, share flagged as signal-rich, share parsed without error. Public-private and locale decomposition.]

D.11 Public competition without log enrollment

[editorial: Carry over from archive. Re-runs the public-side competition regression without log enrollment as a control, for readers concerned that lagged enrollment is itself an outcome of past website-content choices.]

D.12 Tier-1 matched-cell public ontology expansion

[editorial: Carry over from archive. Documents the matched-cell expansion of the public-school LLM ontology sample from 5,591 to 6,078 schools and notes that California public schools are held out of the LLM-based headline regressions pending uniform deep re-crawl.]

D.13 Within-state atom signature

[editorial: Carry over from archive. State-by-state heatmap of within-state mean topic density for the focal topic families. Useful for spotting state-level outliers.]

D.14 Contextual correlates of topic prominence

[editorial: Carry over from archive. Full pairwise correlation matrix of focal topic families with county-level context variables. Companion to Table 13 in the main text.]